

Lab Handbook

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Chapter 1

Joining a lab

No man is an island,
Entire of itself,
Every man is a piece of the continent,
A part of the main.

— John Donne

Although you will be familiar with the names of a handful of scientific heroes, science does not actually advance from the rapid insights of rare geniuses. Scientific knowledge accumulates through the consistent, often painstaking, efforts of groups of people. Across the world, webs of laboratories focusing on related topics work toward the common goal of understanding human memory and communicating how best to use our emerging knowledge to improve peoples' lives. You have opened this manual because you have joined one such group. The purpose of this manual is to help you understand your

role in this endeavor and how to contribute in a way that makes your work maximally useful both to your local colleagues, your international colleagues, and the public, both now and branching into the future.

Joining a lab is about taking what you have to offer - your intellect, your skills, your diligence - and transforming them into new information. This is not just about you working for the lab: the lab also helps you acquire new skills, start your work from a more advanced point than you would be capable of on your own, and eventually boost the signal for transmitting the knowledge you helped create. Everyone contributing to the lab's functioning - undergraduate researchers just starting out, early career researchers, university support staff, the principal investigator - plays a vital role in this effort. To maximize the utility of the work we all do, it is crucial for everyone to:

1. Know where to find information about the lab and the procedures we follow
2. Use the common conventions and procedures we have established
3. Propose and implement improvements to this workflow as new, better methods become available
4. Communicate clearly and openly about our work

Lab procedures have been designed to ensure that the effort you put into your lab work goes as far as possible. Implementing these procedures may require learning something you don't currently know how to do, or doing a task a

bit differently than you have done in another lab. We have examples and support available to help you manage this and learn quickly. No one should need to re-invent anything in this lab that can be copied and modified. You will not be starting your work from scratch, but with at least some building blocks that other researchers have prepared before.

1.1 Hierarchy

1.1.1 Our core team

Labs naturally have a hierarchy. While there is no unimportant work in the lab (it's all important), there is some work that requires a mature knowledge of the literature and a bird's eye view of our international colleagues' latest work. The **PI** (for Principal Investigator) and more senior research assistants are more likely to have this perspective. The most senior staff are also the most permanent, and will know the most about what we have done before, what other labs looking at similar topics are doing right now, and what the most glaring gaps in our knowledge about human memory are. The PI in particular will be the person most likely to be able to see how all the projects in the lab relate to each other and to other contemporary research, to see opportunities for publishing individual pieces of research or combining apparently different strains of research into a single article. The PI's ability to do this well depends on everyone's good documentation of their work and open communication about it. Ensuring that the state of your project is always discoverable to the PI increases the chances that it can form part of

an important communication, which benefits everyone involved.

Post-docs (for post-doctoral researchers) have already earned their PhD, but have not yet started their own lab. Post-docs work here in order to learn more about working memory, get a bit more research experience and more publications, or learn about a method that they did not have the chance to learn about during their post-graduate studies. We are currently trying to secure funding to offer post-doctoral researchers the chance to work on particular projects, but we would also be interested in hosting post-doctoral researchers who want to apply for their own funding to work at Cardiff University on something related to working memory. Interested people should contact Candice Morey (MoreyC@cardiff.ac.uk) to discuss ideas for proposals.

Post-docs are the most senior members of the lab after the PI. Depending on their mission, they might be working on their own project related to the lab's aims, or they might be leading a project that the PI secured funding for. They have experience working in other labs, and can bring their knowledge to our procedures and help us improve them.

Postgraduate Researchers (PGRs) are undertaking lab work as part of their preparation of a postgraduate thesis. PGRs are usually PhD students working for 3 years, or maybe 1 + 3 years if their funding also includes a year of MSc study. PGRs' main priority is completing a coherent body of research for their thesis. All of the research they undertake for that project could end up being published on its own, but pieces could also be integrated into other lab projects. This is one reason cross-project continuity is important.

PGRs will usually move on to a research job elsewhere once their education is complete. At that point, research they carried out that hasn't been published yet may not be their top priority, and the PI and post-docs can be helpful in ensuring that unfinished strands of work continue to flourish.

PGRs may also be paid researchers carrying out designated, funded laboratory work. For most intents, temporary paid PGRs play similar roles to PhD students, but paid PGRs will be expected to work on multiple projects, as hours permit. Paid PGRs will have the chance to be involved deeply in projects, and may have the opportunity to take ownership of a line of research. Unlike PhD students, paid PGR time may be re-directed by the PI to deal with urgent work, like collecting data needed for an invited revision.

Undergraduate Researchers (UGRs) are a diverse group of students with a variety of reasons for joining the lab. Some will join the lab to carry out their final-year project, and will be active lab members from 1 October until 30 April of a single academic year. Others might be visiting students (on Erasmus fellowships or an internship scheme) working for a set period as long as a calendar year. Others might be working for a short period during the summer (e.g., on a Learning and Teaching Academy fellowship). Most UGRs are involved for less than a year, and therefore must be integrated into a larger project. UGRs learn basic research skills working in the lab, and if involved in a self-contained project (like a UG dissertation), they also take responsibility for interim communications about the work they carried out. Because their time in the lab is so limited, it is very important that UG research is well-documented so that the lab can efficiently carry it forward.

1.1.2 How the hierarchy works

The PI, for better or worse, shoulders responsibility for the work conducted by her lab group. While everyone involved in the work will be acknowledged when work we have done is published or praised, the PI will always be primarily responsible for correcting problems when they arise, no matter who really caused them. Our work can be questioned years after it has been carried out and published, meaning the PI is the only person committed to this for long enough to realistically keep this commitment.

For some post-doctoral and PGR projects, the researchers involved might share a long-term commitment to the research and be the “local PI” on the work. In those cases, they will act as the primary person responsible for those projects. Even so, the PI must always have access to enough information about these projects to independently reproduce analyses and replicate findings.

While the PI thinks in terms of large, multi-experiment projects, lab researchers at all levels will have the responsibility for individual experiments, projects, or component projects. Elements of any project must always be documented. Every project has designated milestones at which documentation should be completed, backed-up, and shared (at least with the lab group, often publicly). This standard workflow will be described in more depth later. Highlights:

1. Whenever a project is communicated at a conference or published, the documentation, anonymized data, and reproducible analysis scripts

supporting the communication must go live unless doing so would violate our ethics agreement (which in our lab, should not ordinarily happen).

2. Whenever a lab member moves on from the lab, every project s/he led must be documented and made accessible to the PI on ***OSF*** (for Open Science Framework; osf.io). At the point, if the project is not published, the PI must be given full editing authority along with the former lab member.

The work we do in the lab is the PI's life's work. You can count on her to see where the projects you did that were not publishable on their own may become useful when combined with the output of other projects. In our list of publications, the contributions of UGR and PGR lab members are emphasized. Data sufficient for a research paper can take years to accumulate; in some of these cases, the student co-authors had completed their piece of the project many years before the paper was published. The ongoing work of our lab can help boost your work even after you have moved on, but that is only possible if the remaining lab members can fully understand and reproduce the work you did.

No one can accomplish the work we are striving to do alone. By joining this lab, you are jumping into a big, partially-completed project, whether you realize it or not. Help me make your efforts flourish for this big endeavor and for your career by learning about how this one cog in the grand machinery of cognitive science operates.

1.2 Current lab personnel

Dr. Candice Coker Morey, Principal Investigator

1.2.1 PGRs

Genevieve Lawson, PhD student

Helena Leach, PhD student co-supervised with Dr. Dominic Guitard

Kayleigh Hughes, PhD student

Nuno Sobrinho, visiting PhD student from the University of Porto

1.2.2 UGRs

Libby Howell-Jones, UG placement student

Summer Cox (UGR project)

Hanna Lewis (UGR project)

Mariya Kanevska (UGR project)

Scott McConnachie (UGR project)

Yilmaz Ozyahyalar (UGR project)

Rebecca Underwood (UGR project)

1.3 School of Psychology staff who support us

- IT services (it-servicedesk@cardiff.ac.uk) help with computing equipment needs.
- Research administrators (psych-research-admin@cardiff.ac.uk) help with putting together and submitting grant applications.
- Open Access administrators need to know when a paper has been accepted for publication, and can help facilitate making it OA and paying OA journal fees (Psych-OpenAccess@cardiff.ac.uk)

1.4 Local collaborators and academics interested in memory or related things

Prof. Chris Chambers

Dr. Lisa Evans

Dr. David Greeno

Dr. Dominic Guitard

Dr. Phil Morgan

Dr. Richard Morey

Dr. Teodor Nikolov

1.5 Current and recent international collaborators

Dr. Angela AuBuchon, Boys Town National Research Hospital

Dr. Bonnie Auyeung, University of Edinburgh

Dr. Nic Chevalier, University of Edinburgh

Prof. Nelson Cowan, University of Missouri

Prof. Emily Elliott, Louisiana State University

Dr. Helen Hodgetts, Cardiff Metropolitan University

Prof. Julia Karbach, University of Koblenz-Landau

Dr. Vanessa Loaiza, Colorado State University

Prof. Louise Brown Nicholls, University of Strathclyde

Dr. Hanane Ramzaoui, Louisiana State University

Dr. Marlene Rösner, Otto von Guericke Universität Magdeburg

Dr. Alessandra Souza, University of Porto

Dr. Evie Vergauwe, University of Geneva

1.6 Alumni

Dr. David Greeno (ESRC Fellow)

Dr. Tanya Joseph (PDRA)

Dr. Molly Delooze

Dr Jason Doherty (PGR co-supervisee, University of Edinburgh)

Dr. Jaroslaw Lelonkiewicz (PGR, University of Edinburgh)

Dr. Jonathan Mall (PGR supervisee, Rijksuniversiteit Groningen)

Dr. Teodor Nikolov (PGR supervisee, Cardiff University)

Dr. Aoife O'Mahoney (PGR co-supervisee, Cardiff University)

Dr. Clara Overkott (Visiting PhD student, University of Zurich)

Dr. Edyta Sasin (PGR co-supervisee, Rijksuniversiteit Groningen)

Dr. Florian Sense (PGR co-supervisee & UGR project, Rijksuniversiteit Groningen)

Anthea Allan (MSc project, University of Edinburgh)

Gerdien Boven (MSc project, Rijksuniversiteit Groningen)

Erminia Fiorentino (PGR co-supervisee, University of Edinburgh)

Emily Gwatkin (RA, Cardiff University)

Yuhao He (MSc rotation, Cardiff University)

Margot Holweg (MSc project, Rijksuniversiteit Groningen)

Amy Hughes (MSc rotation, Cardiff University)

Kelsey Jones (MSc rotation, Cardiff University)

Ralitsa Kostova (RA, Cardiff University)

Monica Miron (MSc project, University of Edinburgh)

Lewis Montgomery (MSc project, University of Edinburgh)

Dr. Rob Nijenkamp (MSc project, Rijksuniversiteit Groningen)

Marina Nikoletsopoulou (MSc project, University of Edinburgh)

Pamela Paniccia (MSc student, Cardiff University)

Indira Pliska (MSc project, Rijksuniversiteit Groningen)

Lieke Roetmann (MSc project, University of Edinburgh)

Dr. Karim Rivera-Lares (MSc project, University of Edinburgh)

Madeleine van der Reijden (MSc project, Rijksuniversiteit Groningen)

Gary Wells (MSc project, University of Edinburgh)

Jody Wiggers (MSc project, Rijksuniversiteit Groningen)

Andrea Winkens (MSc project, Rijksuniversiteit Groningen)

Ieva Zeromskaite (MSc project, University of Edinburgh)

Samia Ali (UGR project, Cardiff University)

Amy Andrews (UGR, University of Edinburgh)

Emily Barbour (UGR project, Cardiff University)

Laura Bentley (UGR project, Cardiff University)

Libby Best (UGR project, Cardiff University)

Lydia Beck (UGR, Rijksuniversiteit Groningen)

Dr. Malte Bieler (UGR project, Rijksuniversiteit Groningen)

Rieke Buggenthin (UGR project, Rijksuniversiteit Groningen)

April Carne (CUROP intern, Cardiff University)

Emily Charnaud (UGR project, Cardiff University)

Emma Chubb (UG intern and EPS Bursary winner, Cardiff University)

Sophie Clegg (UG intern, Cardiff University)

Yongqui Cong (UGR project, Rijksuniversiteit Groningen)

Molly Delooze (UGR project, Cardiff University)

Poppy Dunbar-Jones (UGR project, Cardiff University)

Aimee Elliott (UGR project, Cardiff University)

Yuet Fong (UGR project, Cardiff University)

Jone McKeever Garay (UGR project, Cardiff University)

Jenny Graham (UGR project, Cardiff University)

Gita Gudka (UGR project, Cardiff University)

Elan Gwilym (UGR project, Cardiff University)

Amy Hamlyn (UGR project, Cardiff University)

Molly Horne (UGR project, Cardiff University)

Christian Hummeluhr (UGR, Rijksuniversiteit Groningen)

Lydia Jones (UGR project, Cardiff University)

Nia Jones (UG intern, Cardiff University)

Harleen Kaur (UGR project, Cardiff University)

Will King (UGR project, University of Edinburgh)

Andrej Krakau (UGR, Rijksuniversiteit Groningen)

Imogen Lee (UGR project, Cardiff University)

Franziska Lehnert (UGR project, Rijksuniversiteit Groningen)

Nadia Lewis (UGR project, Cardiff University)

Tuomos Lundstrom (UGR, Rijksuniversiteit Groningen)

Robin Macy (UGR project, University of Edinburgh)

Dr. Silvana Mareva (UGR, University of Edinburgh)

Kirsten McDermott (UGR project, Cardiff University)

Leire Martin Mendez (UGR, Rijksuniversiteit Groningen)

George Metaxa (UGR project, Cardiff University)

Mary Millar (UGR project, Cardiff University)

Isra Mohamed (UGR project, Cardiff University)

Dr. Anne-Katrin Muth (UGR project, Rijksuniversiteit Groningen)

Rhiannon Newall (UGR project, Cardiff University)

Muhammed Nor Azam (UGR project, Cardiff University)

Rigkerta Ntisa (UGR project, Cardiff University)

Ceri O'Connell (UGR project, Cardiff University)

Sophia Oelmann (UGR project, Rijksuniversiteit Groningen)

Erin Parry-Bennee (UGR project, Cardiff University)

Viktorija Pratuseviciute (UGR project, University of Edinburgh)

Mindi Price (UGR, Rijksuniversiteit Groningen)

Christina Protopapa (UGR project, Cardiff University)

Nur Ahmad Radzi (UGR project, Cardiff University)

Nelly Rahmede (UGR project, Rijksuniversiteit Groningen)

Eden Rendell (UGR project, Cardiff University)

Amilia Roberts (UGR project, Cardiff University)

Lee Robson (UGR project, University of Edinburgh)

Gintare Siugzdinyte (UGR project, University of Edinburgh)

Teddy Spassova (UGR project, University of Edinburgh)

Dr. Antonia Südkamper (UGR intern, Rijksuniversiteit Groningen)

Peter Thomas (UGR project, Cardiff University)

Abigail White (UGR project, Cardiff University)

Emily Wiegold (UGR project, Cardiff University)

Charlotte Williams (UG intern, visiting Cardiff University from Plymouth University)

Harrison Wilson (UG intern, Cardiff University)

Dr. Michael J. Wolff (UGR project, Rijksuniversiteit Groningen)

Baz Van de Vegte (UGR, Cardiff University)

Berry van den Berg (UGR project, Rijksuniversiteit Groningen)

Yixia Zheng (UGR project, Rijksuniversiteit Groningen)

Chapter 2

Preparing for a stellar project

You may be joining the lab hoping to publish some of your work. We definitely aim for the work we carry out to be good enough to promote to colleagues around the world. We always want our work to be convincing and to make an impact. This section lays out our guiding philosophies for making that happen.

2.1 Choosing a research question

Researchers just beginning often struggle with choosing a good research question. That's no surprise: this is really difficult. There are a lot of factors to consider, both in terms of logistics and potential impact. Under pressure to find a question, the junior researcher might settle for a trivial extension that does not have much theoretical depth (e.g., replicating an interesting finding in a different population). Extensions like that are not necessarily uninter-

esting, but they usually lack theoretical drive (e.g., is there a good reason to think that the finding would differ in that population?). The strongest research questions also lead to one of a small set of possible conclusions, not just did it or did it not work. Ideally, you want to be clear about what the competing predictions are, and develop a design where you could learn something even from finding a null effect.

It is also difficult to estimate how much time a project will take. Even with many years' experience, I cannot do this reliably. Enthusiastically, young researchers starting their first project often come up with huge, impractical ideas that would be impossible to complete under their time and resource constraints.

You should definitely think about what research questions interest you, and you should even think grandly about what you would like to accomplish. But after you have done that, let your PI help. Your PI will have the background knowledge you currently lack to identify ways to refine your ideas and give them more theoretical depth. She will be able to suggest directions you can look into. She will know whether what you are considering is feasible or not.

The ability to hone in on a focused and interesting question will develop as you accumulate knowledge. I definitely couldn't do it as an UG, and I knew at the time I couldn't. To get a start on my honor's research, I asked one of my mentors for help. After listening to Prof. Janet Kistner give an informal presentation at a lab meeting, I approached her about doing an UG dissertation (it was optional at FSU, and most US universities actually). My starting pitch was simply that I found her group's project about the

development of motivation interesting, and that I would like to learn more about it, and did she think there was any possibility to carry it further with an UG dissertation project? I wasn't confident that I could independently identify a good way to contribute, but I could explain why this research appealed to me (it was the clever operationalization and manipulation her group used) and indicate that I wanted to learn more about it. So, you do not need to have a brilliant idea ready - you just need to think about your interests and goals enough to express them to your PI. Then together you will weigh options and refine them.

Shifting to the PGR level, you have often identified your topic before you even arrive on campus. Usually, PGRs have agreed about the broad topic of their project with a potential supervisor before even applying to the program. When you are seeking supervision as a PGR, you should be seeking someone who can evaluate the research you want to do. Never spam PIs from whole departments with your request for supervision. Do not bother cognitive psychologists with your proposal about dentistry, or agricultural business administration, or desire to master the Western blot technique (I've seriously received earnest, seemingly personal emails from prospective students seeking my supervision about all of these topics). Put in the effort to identify potential supervisors who actively do work in the field you are proposing to work in. You are unlikely to even get a response from the others. Even if you do, it will just be confusing, and is unlikely to lead to good supervision.

Assuming you have chosen this lab for good reasons, the first step on arrival at your PGR lab is to discuss your PGR proposal with your supervisor. The

proposals we request for admitting students are short and lack much detail. They might contain good ideas, but they are not really plans of action. First thing: sit down with your PI and flesh out a more thorough and detailed plan.

There are three main principles that I try to work into any research project:

1. Try to think of a design that can offer you more than one way to address your point.
2. Think about minimizing measurement noise.
3. Show all your final work, preferably publicly.

I'll elaborate on the first two points now. There is so much to say about the third point that there is a whole chapter about it later.

2.2 Planning for converging evidence

There are many ways to write an adequate scientific article. Often, scientific articles consist of only one experiment, analyzing one dependent variable. Sometimes a report this simple is striking and clean, and all you wanted to know about that topic. However, usually such a report leaves a lot of unanswered questions. In the best-case versions, the single-experiment paper reports a beautifully designed study, with obvious predictions, clear results, with results coming from multiple dependent measures and all converging on the same answer. I don't believe that these lovely, succinct papers come

from single-shot attempts. I think they come from extensive piloting, and eventually the judicious decision to write a brief paper about the ultimate design rather than write a long paper including every step that led to the ultimate design. While we will never hide the preliminary work that went into a published paper, the ultimate communication we make will reflect the evidence that we are confident about: experiments conducted after piloting, when we are sure we understand how the task is working. Though you should not assume that all the data you collect will be publishable quality, or that you will write a paper about each experiment you conduct, you will save yourself some effort though if you consider in advance what kinds of evidence might help you make a converging case, and how much opportunity to acquire that evidence you can work into your experimental designs.

One important tip is to always measure and record the rawest dependent variables possible. For example, we are always measuring accuracy in our work. With accuracy, you can record whether a given response was correct or incorrect. That might be all you are planning to analyze, but it limits you. It might be interesting to know in what way a wrong response was wrong. You might not realize it at the start of your project, you might later think of a way to use errors to determine between two otherwise vague possibilities. By recording only “right” or “wrong”, you are losing information that might have been useful.

I’ve regretted decisions about this myself. I have a few data sets measuring spatial serial order reconstruction via mouse click responses. My program recorded a lot, but I did not manage to get it to record response times for every mouse click. There are exquisite analyses that can be done with

this information (e.g., Chevalier, James, Wiebe, Nelson, & Espy, 2014) that I unfortunately cannot perform on those data sets (e.g., Morey & Miron, 2016; Morey, Mareva, Lelonkiewicz, & Chevalier, 2018).

Our resources are always limited: there is never as much money, or time, or participant energy as we could spend. Make the most of your resources and record as much as you can about your participants' response. Sometimes it will take extra effort in programming your experiment, but if you get to $N=40$ and then decide you want those data, you will not only have to figure it out, but you will have to begin collecting data again. Figure it out first, or at least weigh the risk of recording less with the PI before beginning data collection.

Sometimes, convergence of evidence will only be possible with multiple experiments. Assuming that we have the resources, we want to write one very convincing paper rather than several papers with evidence that doesn't support a complete argument. This also lets you better control your narrative; writing about one isolated and interesting result, you will need to discuss the myriad possibilities for interpretation that you have not yet ruled out. Putting multiple lines of evidence in one paper means that you can rule out more interpretations, which strengthens the interpretation your data support.

It is unlikely that you will be running an experiment, then publishing it, and repeating that process, so you want to plan for serial data collection, for efficiently processing results, for being ready to collect data during convenient seasons for participants, etc. if you want to write convincing papers that are

bursting with clear evidence.

2.3 Minimizing noise

Another factor that separates the so-so evidence from the convincing evidence is how clear and obvious the effect looks. Of course we will be doing inferential statistics, but ideally these only confirm what is already obvious in your plots. This means you should collect more data than you think you need to detect an effect of a particular size (more on that later). You should also think carefully about all of the irrelevant factors that could affect your participants' responses and do what you can to neutralize them in your experiment's design and protocol. These are things like interruptions (phones beeping in the middle of a task, fire alarms sounding), irritating background noises, fatigue, contingencies that unintentionally reinforce random responding, or individual differences beyond those we normally expect (e.g., participants under the influence of a drug). Much of our lab's standard procedures have been adopted through experience and practice to minimize extraneous measurement error. Adapting your protocol from an existing one helps you benefit from someone else's experiences.

2.4 Illustrations of scope and output of student projects

It can be difficult to imagine what a reasonable project looks like if you haven't done one. It is also difficult to generalize enough to say what a great UG dissertation or PG thesis looks like. I'll try though. Naturally every project is custom, but I've attempted to illustrate a holotype - the modal characteristics of a solid, successful project - for UG and PG projects.

2.4.1 Holotype: UG dissertation timeline and workload

In the UG dissertation scheme at Cardiff University there are milestones built into the system. You must have a working title and abstract after about 3 weeks. There is little time for thinking about what research question you are asking and how you will answer it, because by the end of October you need to be thinking about how you will implement your design.

Start off in week 1 (or earlier if it suits everyone) by meeting with your PI and informally chatting about your interests and what skills you hope to learn during the project. Your PI will have a mental short list of possibilities and can quickly see which ones are most likely to suit you, and how they might be adapted to suit you better. You should leave this meeting with a few ideas and a list of readings.

Your project will be one experiment with at least 30 participants. If you are

2.4. ILLUSTRATIONS OF SCOPE AND OUTPUT OF STUDENT PROJECTS³¹

working collaboratively with other students, you might be running a series together or a larger, more complex design that requires a larger sample. But if you are working independently, count on only having time and resources for a single experiment, with ~30-60 participants (depending on session length).

Once you have settled on a question, find your deadline, work backwards, estimate how much time you want in between major milestones (e.g., getting feedback from your presentation and submitting, writing after finishing your analyses, analyzing your data after collecting them, etc.). You will probably find that you want to have at least begun data collection before winter break. This is also a bit early in terms of when most UG students collect data, so you will have plenty of participants.

This gives you one month to implement your experimental design and develop your protocol. Here is where your PI can help. There will be lots of ready-programmed experiments available from other lab projects. Your PI can help you limit your choice of paradigm to one that the lab already uses in some form, identify modifications you may need to make, and help you see how these modifications fit with what has been done previously. You will learn a little about programming, but unless you already know how, it will be extremely hard to learn to program and create a useful program from scratch in a month. Before collecting data, you will have already:

1. Written an experimental protocol explaining step-by-step *how* to run a session
2. Tested at least one pilot participant to make sure your protocol includes every step needed

3. Gone over the output from your program with one of the senior researchers to make sure it records everything you need to answer your research questions
4. Scheduled a real participant for a time when one of the senior researchers in the lab can supervise and check that your implementation of the protocol is fine
5. Created an Open Science Framework page for your experiment (with the PI as a full-fledged user), uploaded your experimental software, your protocol, paperwork, and a written description of your method and planned analyses

When you return from winter break, you will finish your data collection and begin processing your data. Here is another moment to seek advice from the PI. Our lab manual codifies many useful procedures for processing data, but they require understanding R scripts well enough to copy and modify existing scripts. Your supervisor can help you with this, and direct you to ways to learn how to get specific things you need from the data. You should complete our standard anonymizing steps on your compiled raw data (either on your own or with help from a senior lab member), and then save the resulting data file to your Open Science Framework page. You may now analyze the data and prepare for your March lab meeting presentation, where you will receive feedback on your analyses. After that, you have a few weeks to finish writing.

2.4.2 Holotype: PhD thesis timeline and workload

The PhD thesis is a substantial body of work carried out over three years. A good series of PhD studies should produce at least one substantial journal article. Usually, a single “chapter” of the PhD includes the contents of one published or publishable paper. Such a chapter may contain a single experiment or a series of related experiments. A good PhD thesis in cognitive psychology typically contains at least three empirical chapters, plus an introduction chapter and a concluding chapter.

A PhD thesis must be cohesive. It cannot include chapters about different things, because the introduction and concluding chapters must make clear how the chapters fit together. So, you must stick closely to whatever topic you and your PI agree on when you begin.

It is a good idea to identify tasks that you can complete in parallel so that you are not reactively collecting data, seeing how it went, deciding what to do, and repeating the process. I think it is ideal to jump-start the PhD project by simultaneously working on a related lab project while designing your own first project. This gives you hands-on experience with the lab procedures you will be using so that you can start your own project quickly when it is ready, and potentially involves you in one lab publication besides any that come from your PhD experiments. An ideal timeline could involve starting with related data collection upon arrival, discussing a way to meta-analyze your topic with your PI and working on that alongside data collection, and then using the knowledge acquired from meta-analysis to plan your first experiments. You want to be ready to collect some data in the spring, but to

plan for the bulk of your data collection to occur in the academic Year 2 and the fall of Year 3, when participants are widely available. Ideally, January - September of Year 3 is for writing, knowing that you could probably collect another experiment in the spring semester if you think it would help.

Each empirical chapter should be a self-contained empirical paper (that perhaps has been submitted for publication). You will have been preparing these gradually during Year 2 and the beginning of Year 3, so some of the months reserved for finishing writing will be spent implementing feedback from peer review. Ideally you will also have some time to consider preparing applications for post-doctoral jobs.

2.5 Making the most of your time

Do those timelines sound reasonable? Experience has taught me that there is somehow never enough time. Every hour you can save by relying on existing procedures and pipelines will help you make the most of your effort. You will also naturally find ways to innovate our pipeline as you work, so as others have paid it forward to you by programming experiments you can modify and figuring out useful R functions and scripting processes that you may copy, you will pay it forward by periodically incorporating something you have discovered into our process.

2.6 Workload expectations

There is no getting around the fact that scholarship is competitive, and that no matter how organized and smart you are, you will need to work very hard (and almost certainly sometimes for longer hours than preferred or strictly required) to complete your marked work to the highest standard. Especially at the post-graduate level, the quality of your work is your ticket to your next job. The better your work is, the more options you will have in your career after university. I do not mention this to frighten you into over-working, but rather to encourage you to think about how you want to manage this. There must be a balance between work and rest so that when you are working, you are able to make the most of your time.

Students are expected to treat their studies like a full-time job (e.g., 35-40 working hours per week, with holidays to be taken at convenient times with respect to the work). I encourage every new starter to speak with me about this: I need to know what your plans and habits are, and I will accommodate them to the extent possible. Because I do not believe that everyone works best on the same schedule, I encourage every individual to think about the ideal hours they would like to maintain, and whether they prefer a strict or flexible schedule. If your ideal hours are very unusual or somehow incompatible with the needs of your project, we need to discuss that as soon as possible. In all projects there will undoubtedly be times when hustling is needed, but fortunately these times are usually predictable (e.g., to get data collection started in time for recruiting free student participants, preparing developmental studies to be run during half-term breaks, submitting an ab-

stract for a conference deadline, etc.). Everyone should be aware that when they have worked long hours, they are entitled to take some extra time to rest and recover.

My hours and habits should not be taken as directives by trainees. Though I try to keep weekends for myself and to work 7-8 hours per day, I often work more than that and sometimes work strange hours when I am inspired to or trying to meet a deadline. I feel great relief to get something off my desk, so I may well send an email after typical work hours are over. I'm often traveling to attend conferences or visit other universities, which may be in different time zones (another reason you may get a message from me at an odd time).

I shut my work email off when I'm not working and sometimes also when I am concentrating on writing. I do not receive any auto-notifications to my personal devices about email or Teams messages, ever. I advise everyone to do the same. My assumption is that you will receive messages at some time on every work day when you have explicitly planned to attend to them, not that you will be always on-call. You should not imagine that receiving an email from me means that I expect you to instantly answer it. You are never obliged to answer me at night or on weekends even if you receive something from me after hours.

Chapter 3

Lab meetings and communication

Periodically, our whole group shall meet to discuss our ongoing work, progress, plans, logistics, etc. The content of each meeting will be arranged and communicated ahead of time. All lab members are expected to attend lab meeting if at all possible. Lab members should avoid scheduling anything during lab meeting time.

3.1 Time and place

During term time, meetings take place on most Thursdays, 13:00-14:30. Location varies: see Teams calendar for details.

3.2 Standing agenda

1. Logistics: Is anything broken? Are there scheduling conflicts/difficulties? Programs to troubleshoot?
2. Progress report: PI will ask a selection of members for a progress report. In semester 1, these are typically to guide UG project students as they begin their projects.
3. OSF Review: We'll check whether any milestones have been reached lately
4. Research content (ordered by priority; we won't do all things every meeting): Lab members starting a new study will give their elevator speech and demo their program, lab members with new data will describe it for us (must have a graph!), we will discuss a relevant research paper

3.3 Upcoming dates and extra content

We have a channel on Teams for announcements about interesting conferences. A subset of us are regularly involved in meetings of the Experimental Psychology Society, the European Society for Cognitive Psychology, and the Psychonomic Society.

3.4 What's in it for everyone

You may wonder why we do this. One reason is that it is an efficient way for the PI to get information from everyone. The PI will see lab members on other days and times too, but it is useful to know that there is a specific, repeated occasion when we will regularly see each other.

Another reason is that it is unhealthy for researchers to struggle alone with difficulties. We prioritize trouble-shooting in lab meetings so that anyone experiencing a problem has a regular opportunity to talk about it with a supportive group of people. It is also useful for others to hear about these problems even if they are not directly affected because 1) it normalizes the fact that everyone struggles with logistics and programming sometimes, 2) the same problem may impact you eventually, and by attending to others' concerns in lab meeting, you will at least be familiar with it.

We also need a regular moment to mark the interim successes that occur during research projects that don't merit wide attention. Finishing up pilot-testing, submitting a conference presentation, finding something interesting in an analysis are all moments that call for some celebration, but will only be interesting to like-minded geeks.

Finally, you may not yet personally see how another lab member's work relates to your's, but it does. Hearing about this work and reading papers together builds our common knowledge space and affords opportunities for generating new ideas. I hope these meetings will always be interesting. Inevitably they will sometimes be boring. But please make the effort to attend and participate. Though you may sometimes not see the immediate gain, you

never know when our meetings bring some opportunity for you to improve your project or contribute to someone else's.

3.5 Project-specific meetings

Sub-groups of us shall sometimes meet for specific business that doesn't involve the whole lab. For instance, many lab members are students, and students may have contact hours with the PI related to their course that non-student lab members won't attend. The personnel involved in particular projects will convene *Data Round-up Meetings*, in which we compile, anonymize, exclude ineligible participants from a data set, and securely back-up the data ahead of the analysis.

3.6 Communication between meetings

Questions are bound to arise in between these formal opportunities to meet. One way to get resolution between meetings is to consult our lab's Teams channel. Microsoft Teams is a university-supported space for group chat and organization that we use to replace personal email when possible. There is a general channel that all lab personnel should be able to access, and sub-channels specific to individual projects or topics. In Teams, you can see the previous communications, so you can check if the information you need is already documented there. If not, you can submit your query to whichever channel seems most appropriate. If you need a fast reply from a specific lab

member, you can direct message them or @ them in a discussion and cause a notification to be sent to them that their opinion is required.

Whenever you have a question that could be answered by one of several people, or if there are several people that probably need to know the answer, you should pose it on Teams rather than over email. Like with your email, you can tell Teams when to allow notifications to reach you so that you are not disturbed when you are not working.

UGR project students similarly have a dedicated MS Team for communicating about their project work.

Chapter 4

Seminars at Cardiff University

One of the perks of working at a university is that you are expected to spend some of your time developing your interests and participating in university life. You are encouraged to attend seminars whenever something that interests you is on offer. You may want to keep the times of these regularly-occurring seminars free during term time.

- School of Psychology Colloquium: some Fridays from 16:00-17:00, usually in 5th floor common room of the Tower building
- Cardiff's branch of the UK Reproducibility Network: Join the mailing list (<https://intranet.cardiff.ac.uk/staff/get-involved/groups-and-networks/view/reproducibility-network>) and watch for announcements. (Note -although my name is on this I'm no longer responsible for it and I have no idea whether the current staff running this are planning anything soon.)

- Cognitive Development Group: some Thursday afternoons. If interested, request to join the CogDev group on Teams, currently managed by Rhys Davies (daviesrm19@cardiff.ac.uk).
- CURBIC Seminar: Mondays, 12:00-13:00, CUBRIC Seminar Room 2. Watch for emails, sometimes this meeting takes place from 16:00-17:00 instead
- Cognitive Neuroscience, first Friday of the month, 12:00-13:00, in CUBRIC Seminar Room 2
- CUCHDS (developmental) seminar: Third Thursday of the month, 13:00-14:00, in CUCHDS seminar room (1st floor)

Chapter 5

Student projects we support at Cardiff University

5.1 Postgraduate researchers

5.1.1 PhD projects

We currently consider applications for PhD projects.

Several UK funding agencies (including Wellcome, ESRC, and Leverhulme) offer PhD studentships. These opportunities tend to be student-led: the student is expected to write the proposal and application. Even so, it is a big advantage to have the advice and assistance of the potential supervisor in crafting an application for a PhD scholarship. Dr. Morey will assist excellent prospective students with applications provided that 1) the research proposed aligns closely with her research interests, 2) the student demonstrates

capability to carry out a PhD project (e.g., has a strong prior academic record, strong communication skills, and excellent references from previous mentors), and 3) the prospective student makes contact in enough time to develop and prepare the proposal ahead of the funder’s deadline. (Note: “in time” for something like this is, realistically, at least a full month before anything needs to be submitted.)

The School of Psychology offers a few PhD student places every year, and in some years this lab explicitly seeks PhD candidates through this mechanism. We encourage applicants interested in working memory to apply even if the exact description of the project on offer doesn’t sound quite like what you are after. Staff are required to post an idea for a project to enter the School’s competition for PhD places, but Dr. Morey would usually be willing to change the plan for a talented student who has a great idea. Students who love the project as described will also always be encouraged to augment it and make it their own.

There are other routes to funding a PhD that prospective students may also explore. Cardiff University offers limited funding for international student fee waivers and stipends. Some opportunities are general, apply to students from anywhere in the world, and there are some scholarships available to prospective students from particular countries.

5.1.2 DEdPsy, Educational Psychology

Some of the developmental work we do could serve to support the research training of DEdPsy students. Please get in touch with Dr. Candice Morey if

you are interested in hearing about project opportunities and discussing the educational relevance of our basic memory research.

5.1.3 Taught MSc student projects

We are happy to host students taking part in the **Children's Psychological Disorders**, **MSc Psych**, and **Social Science Research** MSc degrees. Appropriate project suggestions will be available annually upon request. Feel free to get in touch with Candice Morey to discuss ideas.

5.2 Undergraduate researchers

5.2.1 Final year UG projects

Every year we host final year UG projects. A selection of project descriptions are available to all rising final year students every spring. Upon joining the lab, students receive an updated list of opportunities for projects. Note though that some projects require the participation of multiple student researchers. Choice of project is always a function of both logistic constraints (including the areas suitable for Dr. Morey to supervise, given her expertise) and student preference.

5.2.2 Professional placements

In some years, we may have the capacity to offer a professional placement to students on a Psychology with Professional Placement course. These positions will be advertised at eligible institutions. Interested students may enquire if they do not see an advertisement in their programme; we may possibly be able to accommodate a placement student even if we have not advertized.

Our placement will be entirely research-focused. This may not fulfill the requirements of all placement programmes.

5.2.3 LTA and SPRINT summer research projects

Cardiff University and the School of Psychology offer funded summer internships for current students. Interested students should get in touch with Dr. Morey as early as possible in the spring. Project descriptions for funded opportunities will be available via JobShop in the spring.

5.3 Pre-university students

5.3.1 Nuffield summer intern placements

We are willing to offer research experiences to teenagers interested in learning about cognitive and developmental psychology some summers. When we have an opening, we will list it in the relevant places.

Chapter 6

When a new researcher starts

There are several steps that must be completed by new researchers working at Cardiff University. Some of these steps are specific to employees, but many apply to student researchers too. For employees, this list is not meant to replace your employee induction, but rather to ensure that things specific to the labs or research work are in place for you expeditiously.

6.1 Research Integrity Training

The Research Integrity Training module covers research ethics. It involves various slide-show based activities and can be completed asynchronously on Learning Central. There is a quiz to pass at the end, which you may take as many times as needed. Research Integrity Training must be completed before you can be associated with any ethics protocol, and must be refreshed every two years.

You can find the Research Integrity Training under *Organisations* in Learning Central.

6.2 Information Security Training

The Information Security Training module covers your responsibilities for maintaining confidentiality and complying with GDPR. It is available on Learning Central and may be completed asynchronously.

You can find the Research Integrity Training under *Organisations* in Learning Central.

6.3 Location-specific training

Depending on where in the university you are working, there may be location-specific safety training or lab inductions you are required to undertake. If working in CUBRIC, completing the CUBRIC new starters form will spark access to the CUBRIC-specific induction activities.

Chapter 7

Lab spaces we use

We do basic cognitive research with young adults and children as participants. Sometimes we make use of special equipment (e.g., eye trackers, EEG). These needs mean we have a stake in several laboratory spaces within the School of Psychology.

7.1 64 Park Place, room 0.05

This lab is available for use by anyone in the School of Psychology, but in practice it is used mainly by us. The room contains two sound-attenuated chambers with work stations both inside and outside the chambers. Priority for using this space: studies with adult participants in which participants must run singly, and for which noise reduction is very important. (This describes much of our research.)

7.1.1 Access to 64 Park Place 0.05

Request swipe-card access from the PI, who will forward your request to psych-facilities@cardiff.ac.uk.

7.1.2 Security in 64 Park Place 0.05

64 Park Place is only accessible to personnel with swipe-card access. The Cardiff University Film Unit is based there, and some staff involved with Registry have their offices there. During regular working hours, there should always be people around to help in case of an emergency. Note the signs indicating fire escape routes and current first aiders.

As with any building that has restricted access, you should not allow anyone to follow you in when you enter, and you should not admit someone requesting to enter if you do not know who they are and what they are there for.

7.1.3 Booking 64 Park Place 0.05

Whether you are programming your study, pilot-testing your study, or running participants, you need to schedule any work time in 64 Park Place 0.05 or risk being kicked out because another person scheduled it properly.

We maintain an Outlook calendar for this lab so that you may conveniently book it. You may request access to the calendar from the PI. Once granted access, you will be able to view and edit the calendar “64 Park Place 0.05” from your university MS Outlook account.

In general, you may book the lab any time it is apparently free. But do keep in mind that other people may need to use it too, especially during peak participant-running times.

We want to always facilitate efficient use of the space. Please consider the following when booking:

- You should make separate reservations per booth so that we can tell if both or only one booth is in use at a particular time.
- Don't "block-book" the booths to run participants for a long period (i.e., a whole day). You may actually use the room for a long period like a whole day, but make a separate reservation in the calendar for each discrete event (e.g., a 1.5-hour session with a participant). This way, if some of your bookings are not taken or cancelled, you can update the calendar so that the rest of us can see if there are times when the booths might be free.
- Besides the date and time, your reservation should include your name, and information about what you are doing (e.g., "participant", "programming", etc.), and which booth you are reserving. This information helps other users know whether they may be able to step in during the booking, and in the event of a mis-communication about scheduling, helps prioritize.
- If booths somehow become over-committed, priority will go to running scheduled participants over programming or preparing. During peak times, plan to do any programming or preparing somewhere else, and

when you need to use the lab to install your programs and test them, plan to do that during a time that is inconvenient for participant testing (early in the morning is usually good for this).

7.1.4 Peak usage

The lab will be busiest during term times. That's mid October until mid December, mid-January until Easter break, during the regular work day (e.g., 9:30 - 16:30). Whenever possible, do not plan on having access to the lab for programming during peak times. During peak times, the best times to use the lab for preparing a protocol, installing programs, or retrieving data are early in the morning (e.g., before 9:30) and during times when both Year 1s and Year 2s have lectures (consult the Year 1 and Year 2 timetables)

7.1.5 Storage

There are shelves in the lab which you may use to store things that you use while in the lab (e.g., experimental paperwork, notebooks, etc.). Bring some sort of container to keep your things in, and label it with your name. After you complete a study, you should remove the paperwork associated with that study from the lab. Please do not leave personal items in the lab after you are finished working there. When you are no longer collecting data for the lab or working in 0.05 regularly, remove the things you have stored there. The contents of the shelves will be periodically binned to avoid clutter, and anything associated with a former lab member that's left behind will be binned or appropriated by someone else.

7.2 CUCHDS

<https://www.cardiff.ac.uk/research/explore/research-units/centre-for-human-developmental-science>

7.3 Laboratories in CUBRIC

7.3.1 EEG

<http://www.cardiff.ac.uk/cardiff-university-brain-research-imaging-centre/facilities/electroencephalography-labs>

7.3.2 EyeLink

<http://www.cardiff.ac.uk/cardiff-university-brain-research-imaging-centre/facilities/cognitive-testing-labs>

Chapter 8

Working transparently on OSF

In Chapter 1, I described how we are working together on something bigger than any single project. My role is both to facilitate your project while you are completing it and to carry it forward when possible after you are finished with it. To do both of those I things, I need to be able to see your work. I need to know what you have finished (and by extension, what you are currently working on) and I need to be able to find your finished products and comprehend them even if I look them up years after you finished them. That means that you need to curate your work in some place that I can always access. By *curate*, I mean create support that guides me through your work so that I, or any one else, can figure out exactly what you did.

It is a chore to curate the materials and data from a project after it is already finished. You forget quite a lot even in the months between starting and finishing a short project. However, while you are working on the project you are actually doing the work I mean when I say *curating*: you frequently need

to explain to me or other colleagues how your experiment works or what you did to the data. So, what I insist on is that you do this in a permanent way by putting this information on **OSF** (that’s for Open Science Framework), rather than in a private way like jotting it down on your private notebook.

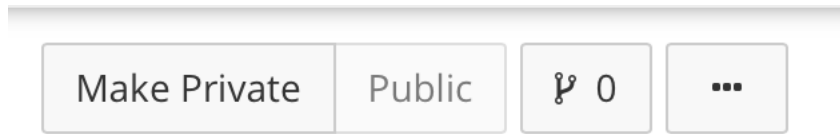
8.1 The OSF environment

OSF is a web-based repository that scientists use to organize their research. OSF is free to use. Sign up for an account on OSF by visiting osf.io and clicking the green **Sign Up** button at the top of the page.

Depending on what you are working on, you may need to create a new project or you may be added as a Contributor to an existing project.

8.1.1 Creating a new project

Most projects we do will share a common structure: they will include directories for preliminary planning, experimenter paperwork, run files, data and analysis scripts, and communications. If your new project will contain most of these, then you can start by searching OSF for the “OSF Research Project Template” designed by Candice Morey. You can look at the descriptions of the components, and if you like, duplicate this template and use it for your new project. To duplicate it, click the button on the upper right side of the screen that looks like a little branch. A menu will appear below.



If you choose “Duplicate this template”, then OSF will create a new project with the same directory structure. You can then fill in the details, delete directories you may not need, or change the names of directories. If you choose “Fork this Project”, OSF will create a new project copying everything about the forked page, including the Wiki content and the project description. All of that can be edited in your new project.

Remember that your project, though new to you, may be embedded with some related ongoing project. If so, it might make more sense for you to be added to that existing project as a contributor. Identifying which circumstance you are in will be one of the first things you and the PI sort out.

8.1.2 Becoming a contributor

One of the lab members may add you as a contributor to an existing OSF page. This will enable you to at least read, and probably read and edit, the existing page. Our projects usually have the same directory structure, but the number of sub-directories in each section may vary depending on how many studies fall under one project heading. If you are added to an existing project, first familiarize yourself with what is there and how it is organized. Then, discuss with one of the other contributors how you should add to the

page. Most likely, there will sub-directories in each folder corresponding to each related project, and you will be asked to create sub-directories for your project.

8.1.3 Naming conventions

For the files you see on OSF, there should be some commonalities in how they are named. You will need to come up with a scheme for naming the files associated with your project. This should not be something too generic like “dissertation consent form.doc” or “experiment 1”: if that were allowed we would have dozens of files with the same names. Pick a short name or acronym for your project that says something about what is unique about it. Then, use that short name to name your files. So, instead of “consent form.pdf” your file might be “wmSpanPrefix_consentForm.pdf”. Note also the use of the underscore: it is good practice never to include spaces in file names. Note also the initial word is lowercase and subsequent words are capitalized. This formatting is called *camelCase*. We frequently have to type file names in contexts where the name must be reproduced exactly for code to run, so it is very helpful if you consistently apply camelCase to all your final names.

8.1.4 Contributors

You can see who all the *contributors* on a project are by clicking the ***Contributors*** button on the upper ribbon. You may be able to add new contributors if you would like for someone else to be able to see your page. You can choose

whether to let a new contributor only read your page or also edit your project page.

At minimum, you and your PI should both be contributors to your project. The PI should always have at least Read + Write access, and possibly Administrator access.

A *bibliographic* contributor is someone who will share in authorship on any presentations or papers that result from the project. If you want to share your page with an outside colleague (maybe as part of a job application, or maybe with a peer who is reviewing your work for you) you can add them as a non-bibliographic contributor. If this person only needs to have a look at your project, then alternatively you can provide them with a view-only link without adding them to the project as a contributor.

8.1.5 Public versus private

Usually, your project will start out as private, and eventually may become public. Our standard is to set a project to “Public” once we have communicated the data outside the lab by presenting it at a conference or submitting a manuscript.

There may occasionally be good reason never to make a project public, for instance if we cannot ensure that our participants’ identities are anonymous. However, normally we can easily do this.

8.1.6 Wikis

OSF allows you to describe each project directory and sub-directory with a Wiki. Take a look at the Wikis on the OSF Research Project Template for an example of what you might typically want to know about the contents of a directory.

8.2 Using your OSF project page to monitor progress

We will both be able to clearly see how your work is progressing from the updates on your OSF page. The pace of progress will vary but the order of events will be pretty consistent. I'll briefly describe the usual project components in the usual order. Each of these steps has its own chapter with more detail later.

8.2.1 First you generate a plan

When we first start discussing the project, we will converge on an experimental design and will work out the possible outcomes we might observe given that design. This may take a while, through a process of reading and discussion. Eventually, you should aim to note down your plans and predictions in a document, and post this document in your Preliminary Plans sub-directory.

8.2. USING YOUR OSF PROJECT PAGE TO MONITOR PROGRESS⁶³

Your Preliminary Plans directory may also include work you did that informed your plan. For example, see the Preliminary Plans directory in *A complex effect of memory load on processing speed*.

8.2.2 Then you prepare your materials

You will need to get, modify, or create software to display stimuli and collect responses, craft a consent form and debriefing statement for your participants, work up an experimental protocol, and test that these materials work to all the contributors' satisfaction.

Once we are happy with the materials (i.e., they have been pilot-tested, the output has been checked, etc.) they go in their appropriate sub-directories.

8.2.3 Then you work on data analysis

Once you have stopped acquiring data, you should compile it into one file (or we may do this together in a *Data Round-up Meeting*), anonymize it (more on that later), and upload it to the Data and Analysis Scripts directory. You can add analyses when you have finished them.

8.2.4 Finally, you communicate your research

When your written report or presentation is complete, you upload it to your Papers and Presentations directory.

At any time during your project, I will be able to see what is on OSF, and with that know what you are finished with, what you should be currently working on. Knowing this will help me give you useful feedback at the appropriate time and be better prepared to help with whatever you might be grappling with.

8.3 Registering milestones

OSF provides free storage for your materials and data. You can think of it as a back-up system that encompasses not just your data, but also your thoughts about your project. That turns out to be really important because we need to keep track of what we believed before seeing the data versus what we believed based on the data.

Making a *registration* creates a frozen, time-stamped version of your project. Registrations cannot be changed, while projects can. Someone could remove data from a project, but not from a registration. It makes sense to register your project at key moments where you might want a permanent record of what you thought before the next phase. Generally, we want to register a project:

1. When we have recorded the definitive study design and predictions
2. After the anonymized data are uploaded
3. When the analyses are finished
4. When the manuscript is submitted (repeat as necessary)
5. When the manuscript is published

These registrations back-up your work, but also allow you to prove when your hypotheses preceded the data.

8.4 When copying is okay

You may be added to an ongoing project, and assume responsibility for some new component of that project. The project might already include materials you could use. But is it a problem to just use them, rather than create your own from scratch?

There are some research materials that it is always okay to copy, or modify slightly for a new purpose. You should always make your consent form from the lab's template. Doing this ensures that important legalese that should be in the form is always there. You do not need to come up with new, original ways to program the same experiment or script the same analysis; it is always fine to copy an existing sample and modify it slightly to meet your needs. You should acknowledge the original creator when doing this, if the original creator is not already a bibliographic contributor on your project (if s/he is, then s/he will already get acknowledged in your work).

You should never copy text destined for your manuscript if you are producing a dissertation or thesis based on the work. Dissertations and theses differ from group-written manuscripts in that a student is composing them to show what they have learned about the work and about writing an empirical report in general. If your project is a dissertation or thesis, you must write your work originally, not quote or refer heavily to the method or result from a

related lab project. You may look at any example documents we have to see what the other authors included, but then close the example and write your own.

If you are working on a project that follows on from another one and is not being submitted for a university degree, then you probably could copy text from related Method sections and modify it as needed. In a serial report of several experiments, the Method and Results of experiments that follow from previous ones are often streamlined by referring back to the first. This is fine for a manuscript intended for publication; in fact, you want to aim for brevity when possible.

Chapter 9

Research ethics

9.1 Basic principles

Scientists have done appalling things for the sake of advancing knowledge, such as intentionally infecting people with diseases, robbing graveyards for corpses to dissect, and leading participants to believe that they had critically injured someone else. To prevent scientists from committing such violations (as well as less dramatically awful ones), international conventions now protect the health, privacy, and autonomy of people participating in scientific research.

Researchers should be familiar with the Declaration of Helsinki and the ethics code of the British Psychological Society. Generally, issues for concern may involve administering medical treatments that might end up causing harm, deceiving the participants, working with participants who are vulnerable and cannot consent on their own behalf, and protecting participants' sensitive

data from actors who may use it for nefarious purposes. We do not carry out any research that involves experimental medical treatment or relies on administration of drugs. We also do not carry out any research that involves deceiving or fooling the participants. We sometimes work with children (who cannot provide consent) and though the data we collect is not obviously “sensitive” or useful to criminals, we acknowledge that we may not be capable of imagining such a use for it, and are therefore under obligation to protect participants’ identities.

The research we carry out does not even approach the inhumane violations that inspired the international conventions on research ethics. Nonetheless there are some principles we must always bear in mind to ensure that our participants are never made to work against their will, never put at risk beyond the risks they experience going about their daily lives, and always receive whatever compensation they were expecting when they agreed to take part in the study.

9.1.1 Informed consent

All adult participants in our studies must explicitly consent to take part in the research. To do this in any meaningful way, they must always be given enough information about the study to understand what participating will be like and an honest assessment of the risks and rewards expected from participating. All participants must be offered this information and must indicate that they accept the main points before they are exposed to our studies.

A template consent form is available from the School of Psychology Research Ethics Committee, available in the Ethics section of <https://inside.psych.cf.ac.uk/>. (Note: InsidePsych is only available to internal users.) The main things that must always be included in the information form:

1. A description of what the participant must do: This is from the participant's perspective. It does not need to say anything about what we are manipulating or why. Withholding that information does not count as "deception". It needs to describe what the session will be like for the participant. Usually, a session will involve seeing some things on a computer screen, trying to remember them, and indicating somehow via computer whether they remember those things or not. The description should be procedural and should not bias the participant in favor of a particular outcome.
2. A statement about any risks and rewards involved: Usually, our studies do not put the participant at any risk beyond what they experience in daily life. They might feel bored or tired during the session. Since those outcomes are generally a part of daily life, they are not considered worth mentioning. Individuals are agreeing to participate voluntarily, and depending on the circumstances, may receive experimental credits toward a course or a small amount of money for completing the session.
3. A statement about what we do with their data: We are committed to transparently sharing all of the data that support our claims with the public. But that does not mean that an individual participant's identity should be publicly exposed. We must tell participants that

we will eventually make the data set they are contributing towards available, but before we do, details that could identify them personally will be removed. The most obvious personal details (e.g., name, student id, address, etc.) we do not usually collect at all, and if we do, we never store them with the experimental data.

4. Finally, participants must be given information about who to contact if they are uncomfortable with any part of the research project. This will be the PI responsible (Dr. Candice Morey) and the administrator of the School Research Ethics Committee (psychethics@cardiff.ac.uk).

9.1.2 Anonymity

Be sure to read the section in this manual about Data Security for a full description of university and lab policy on protecting participants' privacy. The short version is that we do not save "personal" or "identifiable" data, usually at all, and never with the experimental data. It is our responsibility to ensure that we do not save details that could render our data identifiable unless it is absolutely essential, and when we do, that we use a procedure that makes it easy to sever any link between the experimental ids and real-life ids. We apply this procedure as soon as possible, and once it is applied, we can no longer tell which data came from which person. Signed consent forms must be saved for 7 years. The consent forms we save should therefore *never* include the arbitrary id assigned to the participant's data.

9.1.3 Educational debriefing

Whether the participants are part of the university community or from the local community at large, part of the university's mission is to educate. At the end of their session, all participants must be offered information about the research presented in jargon-free language. Participants should also be given information about where they may learn more, and should be given the opportunity to question the researcher working with them or the PI if desired.

9.2 Applying these principles in our lab

Individual researchers will not typically need to acquire ethics approval independently. Most research projects will fall under one of our existing, approved generic headings. Individual researchers will need to ensure that their experimental paperwork, data recording procedures, and protocol are all aligned with these principles:

1. Written consent is always obtained.
2. Consent includes explicit acknowledgement that anonymized data may be shared online.
3. Arbitrary participant ids are never associated with participant's signed consent form.
4. Any document that links personal ids with arbitrary ids must be treated as **confidential**. It cannot be accessible (in digital or print form) to

anyone unnamed on the ethics application. It must be possible to destroy once the links are no longer needed. Once that destruction occurs, it must be truly impossible to re-forge the links. There cannot be a back-up way to reconstruct these links from any recorded data we preserve.

5. An appropriate education debriefing is prepared for every project. Sample debriefings are available in the shared lab OneDrive folder, in the *templateForms* directory.

9.3 Assigning participant identifiers

Assigning an arbitrary id number to each participant is one of the most important bits of lab procedure to get right to maintain data security. This is important both for ensuring that you know which data come from a single unique participant, and for ensuring that all the data you spend time collecting are preserved and usable.

9.3.1 Ids are always unique!

Never re-use a participant number in the same project. Even if you know data will be invalid (e.g., test participant, doesn't meet inclusion criteria). This can cause confusion later, or can over-write files that you may not want to be overwritten.

Always use the same participant number for the same person (e.g., if you are

collecting data at multiple time points from the same person, or on multiple tasks in a session).

Vignette: An ambitious MSc project involved collecting data from participants in two sessions. Participants ran two sessions each, performing two of the four conditions in each session. Eventually though, all of these data needed to be analyzed together. (After all, that is the advantage of going to the effort to get the same data in each participant - you get to do a within-participant analysis.) The student ran each session with a unique identifier. This certainly prevented any data from being accidentally overwritten, but it meant that when it was time to analyze the data, we needed a way to code which ids were actually the same people. There is no elegant way to do this, because there was no regularity between the ids to exploit (coding requires regularity to exploit). Below is the code we used to assign new participant ids to both of the data sets from each participant.

```
# Also, need to link subject numbers. Student ran same participants on unique id num  
# 1=29  
d$Subject[d$Subject==1 | d$Subject==29]=129  
# 2=19  
d$Subject[d$Subject==2 | d$Subject==19]=219  
# 3=16  
d$Subject[d$Subject==3 | d$Subject==16]=316  
# 4=25  
d$Subject[d$Subject==4 | d$Subject==25]=425  
# 5=32
```

```
d$Subject[d$Subject==5 | d$Subject==32]=532
# 6=13
d$Subject[d$Subject==6 | d$Subject==13]=613
# 7=14
d$Subject[d$Subject==7 | d$Subject==14]=714
# 8=24
d$Subject[d$Subject==8 | d$Subject==24]=824
# 9=15
d$Subject[d$Subject==9 | d$Subject==15]=915
# 10=17
d$Subject[d$Subject==10 | d$Subject==17]=1017
# 11=39
d$Subject[d$Subject==11 | d$Subject==39]=1139
# 18=23
d$Subject[d$Subject==18 | d$Subject==23]=1823
# 22=36
d$Subject[d$Subject==22 | d$Subject==36]=2236
# 26=49
d$Subject[d$Subject==26 | d$Subject==49]=2649
# 27=31
d$Subject[d$Subject==27 | d$Subject==31]=2731
# 28=60
d$Subject[d$Subject==28 | d$Subject==60]=2860
# 30=42
d$Subject[d$Subject==30 | d$Subject==42]=3042
```

```
# 33=54
d$Subject[d$Subject==33 | d$Subject==54]=3354
# 34=63
d$Subject[d$Subject==34 | d$Subject==63]=3463
# 35=40
d$Subject[d$Subject==35 | d$Subject==40]=3540
# 37=46
d$Subject[d$Subject==37 | d$Subject==46]=3746
# 38=48
d$Subject[d$Subject==38 | d$Subject==48]=3848
# 41=65
d$Subject[d$Subject==41 | d$Subject==65]=4165
# 43=59
d$Subject[d$Subject==43 | d$Subject==59]=4359
# 44=75
d$Subject[d$Subject==44 | d$Subject==75]=4475
# 45=52
d$Subject[d$Subject==45 | d$Subject==52]=4552
# 47=78
d$Subject[d$Subject==47 | d$Subject==78]=4778
# 50=57
d$Subject[d$Subject==50 | d$Subject==57]=5057
# 51=69
d$Subject[d$Subject==51 | d$Subject==69]=5169
# 53=55
```

```
d$Subject[d$Subject==53 | d$Subject==55]=5355
# 56=66
d$Subject[d$Subject==56 | d$Subject==66]=5666
# 58=67
d$Subject[d$Subject==58 | d$Subject==67]=5867
# 61=64
d$Subject[d$Subject==61 | d$Subject==64]=6164
# 62=71
d$Subject[d$Subject==62 | d$Subject==71]=6271
# 68=74
d$Subject[d$Subject==68 | d$Subject==74]=6874
# 70=72
d$Subject[d$Subject==70 | d$Subject==72]=7072
# 73=76
d$Subject[d$Subject==73 | d$Subject==76]=7376
```

While this code solved the immediate problem, it was a clunky solution that could have introduced many new errors that would be difficult to detect. What if the student lost his notes indicating which participant ids should be paired before running the analysis? What if those notes contained discrepancies he could not sort out (e.g., writing down the same id accidentally for two participants)?

Furthermore, the inelegant coding solution needed to manage this is also prone to human error. Mis-typing an id number could mean creation of a bogus pairing, accidentally assigning three sessions to one person, etc. After

doing this, we had to check carefully that all resulting ids had the right number of sessions, trials, and conditions.

In the end, this worked out fine. But work and hassle would have been avoided if each participant were given a consistent id to start with, or alternatively if an id system was used that resulted in unique session ids that were related in some regular way. This problem would of course have been compounded if the design had been more complicated (e.g., what if there were > 2 sessions?) and as the sample size increased (imagine sorting this for 300 participants instead of ~ 30).

9.3.2 Ids are always arbitrary, anonymous

The system you choose for assigning participant numbers should not preserve any links between the participant's identity in real-life and their identity as a unique contributor to your data set. Ids should not be derived from any university-linked identifier, address, or birthdate. They should not include the date or time of the session. Typical ways to assign ids are to choose a starting number and increment upward with each new participant.

If you are running a multi-experiment project, it is a good idea to segregate the participant ids of subsequent experiments in some way that makes them obviously distinct from previous experiments in the sequence (e.g., experiment 1 starts from 100, experiment 2 from 200, etc.). This way, if you ever combine data to perform a multi-experiment analysis, you still have unique participant ids.

9.4 Ethics approvals currently in place

9.4.1 Limits of memory (EC.17.09.12.4952G)

This is a generic approval, and now historic. It covers several projects with data collection taking place through summer 2024. This heading covers behavioural studies with adults (17-40 years old) in which participants are asked to try to remember something, sometimes while also doing something else. This heading currently applies to single-session studies of 90 minutes or less.

9.4.2 Memory storage and processing across the lifespan (EC.18.09.185339GR)

This is a generic approval valid until 31 December 2025. This heading covers behavioural studies with adults and children, eye-tracking, and multi-session data collection. Examples of studies under this heading include manipulations of complex span tasks and the Flavell et al Registered Replication Report.

9.4.3 Incidental memory (EC.23.05.16.6803GR)

This heading covers the paradigms where we unexpectedly ask for memory of some feature that was not originally instructed to be remembered. This includes our plans for an external many labs study and also covers EEG data acquisition.

9.4.4 Working memory limits (EC.24.09.10.7051G)

This is a generic approval valid until 31 December 2044. It covers any short-term memory study with consenting adults in which we do not need to record any personal data and show only materials that are not upsetting. Data collection can be online or in-person, and we can collect eye tracking or EEG data.

Chapter 10

Ethics at Cardiff School of Psychology

Ethical approval is an essential part of conducting research. At Cardiff University, each academic school has its own School Research Ethics committee (SREC) that is responsible for ensuring that all research carried out within the School has been subject to an ethical review. The university's procedure for ethical review allows staff and students to achieve research excellence and produce work with the highest standards of research integrity and professionalism.

10.1 Who needs to apply?

Staff and students conducting research on humans must apply for ethical approval. In some cases, ethical approval may have been granted by an

external organisation (e.g., NHS) and this approval letter must be submitted to the SREC.

You may not need a completely new application if your research project fits under one of the lab approvals already in place. Speak with your supervisor early in your project to determine what you must do to ensure your work is covered by ethics approval. Even if you do not need to make a new application to the ethics committee, you likely need to complete some training and have a conversation about ethics with the staff member supervising your research before you are allowed to begin.

10.2 How to apply for ethical review?

A great starting point for information and resources about the ethical review process can be found at InsidePsych. Log in with your Cardiff University credentials and navigate to the ***Ethics*** tab which can be found in the Undergraduate, Postgraduate, and Staff pages.

10.3 Mandatory Research Integrity Training

All researchers must complete the Cardiff University Research Integrity Training module. PGRs and staff members should automatically find this as an enrollment on their Learning Central page. UGRs can access it here.

10.3.1 Types of applications

All applications and supporting documents must be emailed to the Ethics committee at psychethics@cardiff.ac.uk

1. Proportionate applications: This procedure is used for proposals that do not raise any ethical issues and recruits participants from the School's participant panel. To recruit from the panel, you will need to sign up for a researcher account on the School of Psychology's Experimental Management System and obtain ethical approval before you begin recruiting participants. Proportionate applications can also be used to make minor amendments to a project that has been previously approved. These applications must be submitted by midday on a Friday and a decision will be made within 7-10 days.

2. Full applications: These applications require more detailed descriptions of the experimental procedures, research design, and ethical considerations. Full applications must be submitted at least two weeks before the Ethics Committee's next meeting. The committee usually meets on the first Tuesday of every month.

Full applications are used when a research project involves (a) recruiting special groups of participants (e.g., children, people in custody, those lacking capacity to give consent, or people engaged in illegal activities); (b) highly sensitive topics (e.g., domestic violence, terrorism, etc), (c) the use of a drug (e.g., tobacco, caffeine, or alcohol), (d) the collection or use of human tissue, or (e) the preservation of personal data (e.g., personal identifiable information stored with data, voice or image recordings, etc.). In essence,

if the research poses an ethical consideration, a full application must be submitted for review.

3. Generic applications: Sometimes, researchers may have a range of studies they would like to conduct within a research paradigm. In these cases, a generic application can be completed so long as there is sufficient description of the different studies they wish to undertake.

For example, a researcher may want to study the effectiveness of recall in different situations. A generic application can be submitted so long as the different studies within this paradigm are described, for example, testing memory in a noisy environment or after participants have had a cup of coffee. Depending on the ethical considerations in the study, these applications may either be a fast track or full application. **Undergraduate students are not permitted to apply for generic approval.**

4. Amendments to approved projects: Sometimes, changes may be made to a project that already has ethical approval. If they are changes to the research team or the project end date then send an email to the Ethics Committee with the details of the change and the project's ethics reference number. All other changes require the submission of a new ethics proforma along with the previous ethics reference number for the project. Remember to provide sufficient detail about the changes as if they are minor amendments they will be dealt with through the fast track route while if they are major changes they will be considered in the same way as a full proposal.

10.4 What forms will I need to complete?

10.4.1 Ethics proforma

An ethics proforma must be completed for all applications except those seeking approval for minor amendments. The proforma can be downloaded [here](#). You will also find guidance notes for the proforma on this page. The proforma asks for details about your study - who is conducting it, who are the participants, and what methods will be employed. If you are conducting a project with no foreseeable ethical considerations then your application will be considered under the proportionate route. If you are conducting a project that raises ethical considerations then you will need to include additional details and materials for review.

10.4.2 Risk assessment

Risk assessments are a mandatory part of the ethical review process. When you fill in the proforma, you will also be prompted to complete an online risk assessment form. There may already be a risk assessment in place for the lab facility you are using and if your project does not add any new procedure to that assessment, you might be able to use the existing risk assessment number for your proposal. Speak with the manager of the lab space to determine whether your project requires a fresh risk assessment. Even if it does not, you must familiarize yourself with the existing risk assessment.

If you do need to complete a new risk assessment, there is a link in the proforma to the online instrument that begins this process. This instrument

is designed to provide an overall estimate of the risk involved in the project and who is likely to be affected. You will be asked to rate, on a scale of 1 to 5, the likelihood of a hazard causing a problem and the severity of injury resulting from the hazard. For example, equipment like computers may pose a risk of electric shock, however, with control measures like PAT testing the likelihood of electric shocks may be low (rated as 1 - very unlikely) while the severity may be rated as 3 to indicate that an electric shock may lead to some temporary injuries. Once you have completed and submitted the form, you will receive an email to indicate if a full risk assessment is needed. If so, the email will also contain a link for the full risk assessment form, guidance notes, and details for where to send the completed form. You will have to give information about the hazard, measures to mitigate its impact, and ratings for the likelihood and severity of the hazard. **Please note you cannot submit your ethics application to the committee before you have a valid risk assessment number.** When planning an application that requires a new risk assessment, assume that you will be asked to revise your risk assessment based on feedback, which may take a few extra days. Once your risk assessment is approved, you will receive a unique *Risk assessment receipt number* via email. You must include this receipt number in your Ethics Proforma when you submit (your application will be returned to you if this information is not included).

You can now submit your ethics proforma together with any supporting documents for ethical review! **Remember, you CANNOT begin any research activities with participants until your application has been approved by the Ethics Committee.**

Chapter 11

Data security

Data security refers to the principles followed to protect our data. There are two sides to this: ensuring that our data are not lost and ensuring that aspects of our data that are personal are kept confidential.

All lab personnel should complete the web-based Data Security training module offered by Cardiff University.

The data we post publicly online should not afford discovery of participants' identities. There nonetheless remains some information in the data file that interested parties could use in conjunction with other known information to identify a participant. Suppose you are a private investigator, and your client wants you to spy on her child at university. You know your mark is enrolled on the Psychology degree, and you find out that completing this degree involves participating in laboratory studies. You think these studies might be a source of key information about how well your mark can remember abstract visual images, so you follow him to the laboratory, noting the date

and time he participated. Once the data are posted online, you can download them and find the participant whose data were recorded on that date and time, even if no other information in the data uniquely identifies your mark. Knowing what we know about the distortibility of data, we can take those visual recognition task data and dubiously infer all sorts of maliciousness.

This is obviously ridiculous, but technically possible. It is referred to as the *motivated intruder test*, and we cannot consider our data truly anonymous if a motivated intruder could easily uncover and exploit the indirect links between anonymous data and identifiable information we may have access to. To comply with EU privacy regulations, we need to take steps to protect participant identities even when we think our data are harmless. This holds regardless of whether we plan to make the anonymous data set publicly available. Unless you are storing the data only on secured Cardiff University resources, it must be properly anonymized. This means that before we anonymize it, data cannot be held even temporarily on personal disks, on DropBox, or Google Docs, and cannot be emailed to non-university email addresses.

11.1 Backing up digital data

11.1.1 During data collection

All lab personnel will be given access to a OneDrive folder “lab-BackUp_cmcorey”. While data collection is ongoing, the researcher responsible for data collection should add newly acquired data to the

OneDrive lab back-up every day new data are acquired. There are sub-directories in *labBackUp_cmorey* for all the physical spaces we use to collect data. Simply copy the your experiment directory on the physical machine to the corresponding sub-directory in *labBackUp_cmorey*.

Note that here it will be necessary to make sure that your experimental directory has a unique name. The short-hand name you use for your project can be used for this. Do not choose names like “project”, “dissertation”, “thesis”, “experiment 1”.

These data are *confidential* because they include timestamps that, if linked to the participant’s EMS schedule or to information the participant discloses publicly about their participation in experiments, their identity could be reconstructed. You should not save these data onto portable drives - they are not secure enough. They should be saved only on the *labBackUp_cmorey* OneDrive, to the lab computers which sit in a secured location, or to password-protected computers associated with Cardiff University.

11.1.2 When data collection is closed

Once we have decided not to collect any more data on a project, it is time to compile and anonymize that data set. This should always occur before any analyses are attempted, and must always be done in a scripted fashion. For lab members not fluent in scripting, this should be done in a face-to-face meeting with the PI or another senior lab member. This is the **Data Round-up Meeting**.

11.1.3 Cleaning up lab directories

After we have compiled the data, created anonymized versions, and backed up the original data to Dr. Morey's H drive, it should not matter whether the original data files stored on *labBackUp_cmores* or on local machines are lost. These sources are sufficiently secure for our data, which are at most *confidential*, never *highly confidential*. In the event that we collect *highly confidential* data, such data should be deleted from local machines and OneDrive as soon as we have the compiled data backed up to the more secure H drive. For data that is *confidential* we need not rush to delete it. We shall allow most data to exist on *labBackUp_cmores* until the space is required for newer data, and then delete the oldest files that have been compiled and backed up first. The researcher responsible for the data should delete it from the local machine where it was collected after data round-up for that project is finished.

11.1.4 Summary

Individual data files may be found on the local machine on which they were collected, unless there has been a crash or overwriting error. Assuming our lab back up procedure was followed, individual data files should be redundantly available on our shared OneDrive back-up.

Once we have compiled and anonymized the data, it will not be necessary to access the individual data files. Lab personnel associated with a project shall all have access to the anonymized data via our shared OSF project page.

This anonymized data file may be saved redundantly anywhere. It will not contain sensitive or confidential information.

If you somehow lose the anonymized data file and it is somehow gone from OSF, we can recover it by re-running the anonymization script we created on the original compiled data. The PI controls access to the original data, so requests for recovery from the original data must go through her. If we need the original data to prove something about the time period of data collection, the PI must also arrange access.

11.2 Processing paperwork

11.2.1 Consent forms

Hopefully, the only paper associated with your project is the signed consent forms. These are definitely identifiable - they contain the participants' names and dates they took part. We must keep copies of these because it is necessary that we be able to prove that a particular person consented to take part in our research, if the person ever claimed s/he did not give consent.

Once data collection on your project is closed, you must turn these papers over to the PI. First, scan the consent forms into one .pdf document. Enter the PI's email address as recipient. Make sure this is done before the compilation meeting. During the compilation meeting, we shall check that the number of pages in the .pdf consent scan matches the number of participants in the study. If not, we may need to discover which participant did not consent, and remove that participant's data using the timestamp information.

Dr. Morey will save the consents .pdf to her H drive. The paper consents will be destroyed in the next confidential waste pick-up.

The consents sub-directory on Dr. Morey's H-drive is organized by year. We are required to keep the signed consent forms for 7 years. Annually, she shall delete the sub-directories more than 7 years old.

11.2.2 Paper data

If your study involves data collection on paper forms, you must back these up digitally by scanning them to .pdf as soon as your data collection closes. While collecting data, you should leave the paper forms in an organized space within the lab.

Paper data should also exclude personally identifiable information. It will also be considered *confidential* because some combination of lab personnel could work out the identity of the participant by linking the participant id on the paperwork with corresponding digital data with a timestamp. This is why paper data should remain in the secure lab. Once scanned, the digital file shall be saved to the appropriate OneDrive directory and to Dr. Morey's H drive.

Data requiring transcription or coding must be duplicated by two independent coders, with any disagreements resolved by a third person. For projects involving coding or transcription, once we have multiple ratings the raters and PI (or a senior researcher) shall meet to compare the ratings and arbitrate any differences. In this meeting, the two rater's transcriptions shall be merged, a resulting measure evaluated, and ultimately the resulting agreed

measure shall be merged with the other project data. All steps will be documented via script. Once this has been completed, and the data have been uploaded to OSF (for anonymized data) or Dr. Morey's H-drive (for non-anonymous data), the paper forms shall be destroyed via Confidential Waste collection.

11.3 Confidentiality

We have procedures built into our back-up process to prevent potentially-identifiable research data from leaving the secure university environment. However, our biggest efforts for solving this problem are taken during data collection and study design. Whenever possible (which is nearly always), we do not record any identifiable information about our participants. Unless you need precise information about their ages at test, we do not record birth dates. We never record proper names or any stable identification number with experimental data. When we need to link a participant's data across multiple sessions, we will do this with a key of real-life identity and contact information to participant numbers, which is destroyed once we have finished collecting data. When we need to use such a key, it is maintained by one researcher, and stored within the secured university environment.

Our ethics approval states that we shall destroy this key when data collection is finished. For projects involving a key, key destruction will be the last item on the agenda of the **Data Round-up Meeting**. Once we have compiled and written the anonymous data files, we shall run tests to be sure that we have correctly matched up participant ids across sessions. Once satisfied of

this, we shall delete all copies of the key.

Chapter 12

Designing your study

Investing time into designing your study pays off in the long run. Testing participants is tedious, hard work. During some periods and for some kinds of participants, it is expensive. You want to be as sure as possible that the experiments you run are capable of addressing the points that you are trying to address. What's more, you want your experiments to address these points thoroughly and well.

We always try to design experiments according to the following principles, unless we have a very good reason to deviate from them.

12.1 Use computers whenever possible to control stimulus presentation and response collection

Back in the late 1990s when I was an undergraduate student interested in cognitive psychology, I worked in several labs trying to figure out what kind of research I wanted to specialize in. One thing became crystal-clear with experience, and it is even more true now than it was then: if you are serious about becoming a cognitive psychologist, you must be proficient at getting computers to work for you. Without decent tech skills, your work will always be sub-par. Yes, some classic papers were written in which the researchers presented their stimuli in low-tech ways, and the quality of the results were based on their great ideas and commitment to conscientiousness. But even if you have a comparable commitment to conscientiousness and comparably great ideas, there will be someone else who has those things and also has sufficiently good technical skills to automate their research designs. And in 99 cases out of 100, that automation will improve the project. Why is that?

12.1.1 People make more mistakes

Suppose you want to show participants a stimulus for 1 second. For our research, it may not be extremely important whether the participant is exposed to the stimulus for 900 ms versus 1200 vs vs 1000 ms, the fluctuations in time you might expect if a human is reading out stimuli or manually displaying them on flashcards. But you don't really know, do you? Suppose a

12.1. USE COMPUTERS WHENEVER POSSIBLE TO CONTROL STIMULUS PRESENTATION A.

reviewer asks for some confirmation about timing, or thinks that a particular pattern in the data might make sense for stimuli presented for 1100 ms, but would be surprising if they were presented for < 1000 ms. You will need to be reasonably certain about major elements of your methods, and reasonable certainty won't be possible with human-manual presentation.

Computers have brief hiccups that can cause events to be displaced from their stated timeline slightly. These displacements are usually on the order of 7 ms, and these blips are recorded. People deviate way more. Suppose you are showing a participant stimuli on a series of flashcards, aiming for exposures of 1 second each, and you need to sneeze. There is going to be an irregularity of way more than 7 ms. What's more, you will have to decide, in real-time, whether to record the presence of that regularity in the trial somewhere, which will again take more than 7 ms. If you decide not to, then we will not know from the data that the irregularity occurred, and that trial may be analyzed as though it were normal. There isn't a great solution for this. But with a good experimental presentation program, we would always be able to check whether our assumptions about timing were met, and if not, how much deviation there was. We could quickly exclude any trial with massive irregularities if we decided it was important to do so.

Another example: people are notoriously terrible at judging randomness. If we want some variable to be presented in random order, a human will have trouble distributing the levels of that variable in a random way. What's more, different humans will have somewhat different ideas about which patterns look random and which don't. It's better to let machines sort this out, based on some rules we restrict them with.

12.1.2 People are biased

The first rule of objective research is that your hypothesis should not be allowed to affect the outcome of a test. This becomes impossible to regulate when the human delivering the stimuli and recording the responses has a hypothesis about how participants will perform under various conditions. Effects of experimenter bias on test outcomes have been demonstrated in many settings. An example: Tryon ran a research program in the 1930s and 1940s aiming to test genetic inheritance of intelligence in rats. He tested rats on completing a maze, and segregated the rats into groups based on how well they got through the maze. He let the ones with the fewest errors interbreed and the ones with the most errors interbreed, and tested their offspring on mazes for generations. The rats in the “bright” population got better and better than the rats in the “dull” populations generations after the first selection occurred.

This may not be so surprising: nearly everyone believes that intelligence is at least partially genetic. But there was more to Tryon’s measure than just that smart rats breeding with other smart rats produces smart baby rats. Tryon’s studies were not blind. The researchers working with the rats knew which ones were “bright” and which were “dull”. Rosenthal and Kode (1963) replicated the research design, but randomly assigned rats from the same population to “bright” and “dull” groups. There were in fact no systematic differences between their “bright” and “dull” rats, except that the researchers working with them believed that there were. Rosenthal and Kode found the same effects as Tryon: the “bright” rats produced offspring

12.1. USE COMPUTERS WHENEVER POSSIBLE TO CONTROL STIMULUS PRESENTATION AND RECORDING.

better at completing mazes than the “dull” rats.

One can imagine how this played out, and the many sources where bias that might affect the rats behavior itself or even just the recordings could be introduced. The researchers would have cared for the rats, making sure they had food, water, stimulation, and clean bedding. The researchers would have put the rats in the maze, counted their errors and timed their performance. The “bright” rats might have been petted more, had their food and water refilled more often or more quickly. When being put in the maze, there could have been differences in judging what was an error, or systematic differences in how quickly the stop watch was triggered and stopped. If you expect the “bright” rat to go very fast, perhaps you very attentive to its run, and extra vigilant about stopping the timer.

This bias isn’t restricted to rats. It can happen any time a person has to make a judgment, however minor, about what a participant did, or make a decision about when to deliver a stimulus, stop a clock, record an error, etc. Can this be counteracted? We could take care to always ensure data collectors do not know the hypothesis. But then who would collect data? Usually, students collect data for their own project, which they have contributed to developing. They can hardly keep themselves from forming expectations about the results. The best and most practical way to restrict experimenter bias is to take as many decisions as possible out of the hands of the human experimenters during data collection.

12.1.3 It's more work in the end *not* to automate

All of our data will eventually need to be quantitatively analyzed. Even data that may begin as qualitative, like responses to questions about how participants performed a task, would eventually need to be coded as falling into one response category or another for some kind of quantitative analysis. So, if you plan to have participants manually write responses, or if you code their spoken responses in real time by hand, you will still need to transfer all of that data to a .csv file. Errors will certainly be introduced when transcribing, so it would not be enough for one person to transcribe these data: it would need to be done independently by two people, compared, and then discrepancies would need to be transcribed again.

Properly programmed and tested, the computer cannot mistake which button the participant pressed, or where the participant clicked the mouse, or what the participant typed. The computer's "judgments" need to be checked at the start, to be sure we know it is systematically recording what we meant for it to record, but once we are satisfied, we can be confident that it is recording these responses the same way every time. It will not get tired and make random errors. It will not forget which button is which. The responses it records do not usually need the time-consuming checking that human-coded data need. A little time spent upfront will therefore save you hours of painstaking work later.

12.2 Work from a good example

Whether you are a skilled programmer or totally new to it, your first step to programming your study should be to find an example program that does part of what you need, and work from that. Even though we are currently switching from closed-source E-Prime to open-source PsychoPy, we already have programs or programs-in-progress that execute spatial and verbal memory span tasks, display spatial locations around a circular perimeter, and administer complex span. Don't start from nothing: break down what you need for your program to do, and ask other lab members for advice finding examples.

Examples can come from inside or outside the lab, but if you are programming something that is incrementally different from something we already have, you should work from the lab version. This is important because we want to make sure that different iterations of our studies differ only in the ways we are aware of. If, when exploring programs from other labs, you discover a better way of implementing something, put that on the agenda for the next lab meeting, and explain what we would gain from adopting the other method. Then 1) we all learn something new, 2) we decide as a group whether the method you discovered is worth implementing across projects, 3) we are all at least aware if one task iteration is going to differ in implementation from other similar implementations.

Another reason to work from an in-lab example when implementing your project is that it increases the likelihood that programming conventions some of us may have previously agreed upon persist in new programs. It is always

rough figuring out just what someone else’s code does. The platforms we favor have some features that help with this, and by working from an in-lab example, you may be able to continue in the style that the rest of us understand.

Most importantly, we don’t have a formal ban on closed-source software. We may sometimes use E-Prime, MatLab, or whatever when it is convenient to do so. Good reasons for this include: carrying on from existing projects that use those platforms (e.g., avoiding “changing horses mid-stream”), or when we are collaborating with colleagues who insist on these (though we will generally try to persuade them to let us provide open-source materials). If you are starting a new project, assume that you will be using open-source software. We do not want to use closed-source software indefinitely, and starting anything new with it perpetuates its use. Consider (especially if you are a student) that your next job may not include access to E-Prime or MatLab, so learning an open-source solution makes it more likely that you can continue working with the platform you become familiar with later, rather than being forced by a new employer to learn something else.

We favor open-source programming platforms that make use of some sort of graphical user interface, like PsychoPy and OpenSesame. Even if you are an expert coder, you should plan to implement as much as possible using the GUI framework. This is because our lab will never exclusively feature expert coders who can follow along with your code. Our group will always include students who are brand-new to programming and coding, and we need our general-purpose materials to be comprehensible to new users who may wish to use them as a starting point later. Many (maybe most?) of our programs

will include some code components, but when it is possible to build in the interface, that's what we do. Because we need for newbies to comprehend things, code components should be obsessively commented.

12.3 Record as much as you can

Though we are committed to documenting our hypotheses and analysis pipelines before collecting data, it is inevitable that we will sometimes think of a great way to test a hypothesis later. There is nothing inherently wrong with data-driven analysis, so long as we are honest about what we predicted before seeing outcomes, so we should try to facilitate getting as much out of each precious data collection opportunity as possible. Programmers must of course ensure that the dependent variables needed for planned analyses are recorded properly, but they should also consider whether other elements of the response may also be conveniently collected. Examples:

1. Your main interest may be response accuracies, but that should not prevent you from also collecting response speeds (and checking that you know how your program records them)
2. Your main dependent variable might be whether a whole sequential response is 100% correct or not, but that should not prevent you from collecting the components of the response (e.g., that the participant responded 8675309, not just that they got a list “right” or “wrong”).

There could be good reasons to aurally record spoken responses, to have response speeds for individual elements in a list response, for elements I've

not yet imagined. You should record as much as you ethically can, and try to think in advance of the ways in which various response elements would work together to converge on a hypothesis (and include those in your preregistration when possible). If you want an element later and did not record it, the only solution is to collect data again, which costs time and (sometimes) money.

12.4 There are always exceptions

There will sometimes be projects for which we don't follow these principles. Sometimes, we will have a good reason for wanting participants to give natural, vocal responses, which currently cannot be evaluated in real-time by a computer. There are bound to be other exceptions. Expedience is **not** a good reason though. If you have no idea how to implement something, ask for the support of the group.

Chapter 13

Should I run my study online?

Lab work can be grueling. It can take many hours of effort to collect data in-person. Sometimes this effort is necessary; for some paradigms, we need to be certain that participants were focused on what they were meant to do, or used a certain standard of input device or speakers/headphones, or had the lights on to a certain level.

For some experiments, we do not necessarily need this degree of supervision, or can easily tell whether participants engaged from their performance on the task. In these cases, we can collect some or all of the data online. Unless you are tech-savvy and familiar with how web browsers work, collecting data online may not save you effort because the learning curve to set up your study will be steeper. You need to learn a lot about this to ensure that your participants have a good experience and that you know what they are seeing and doing. But if you can set this up, you can save a lot of time compared with collecting data in the lab.

You should always talk through the pros and cons of in-person and online data collection with the other members of your team at the start of a project. There are likely resources you can work from if you decide to go for online data collection.

13.1 Key questions to ask yourself

1. In my paradigm, are there plausible checks I can build in to ensure that participants engage with my task? Options include a practice session with feedback, or possibly requiring a minimum level of accuracy before a participant can proceed past practice.
2. Will I be able to tell the difference between an engaged participant and a random responder? Are there checks I can include to test for random responding?

See this useful advice from Prof. Jenni Rodd about ensuring data quality when running participants online.

If you cannot be sure about your data's quality if you run online, then you should plan to run in the lab.

Chapter 14

Running studies online

14.1 General principles

To recruit online, you will need access to participants, and you will need to have hosted your experiment on a server. We have access to multiple platforms for these.

14.1.1 Participants

We use SONA systems, locally called Experiment Management System or EMS, to recruit students as research participants. Participants may be recruited via EMS for in-person or virtual sessions.

We can recruit paid participants via Prolific (which we prefer to other options for running studies online; see Uittenhove et al.'s paper about this).

14.1.2 Servers

Cardiff University pays for unlimited access to pavlovvia.org. We can serve experiments from there. This is especially convenient when moving a study created with PsychoPy Builder online.

We can also host on mindprobe.eu, a freely available JATOS server supported by ESCoP. This is an especially convenient option when working with studies created OpenSesame. To use mindprobe.eu, you must request an account. Allow some days for this process.

14.2 Automated communication between server and participant recruitment platform

You need to allow for your experimental program to collect a static participant id from your recruitment platform, and also to send the participant back to their recruitment platform for completion to be acknowledged. This magic happens via the URLs provided. There is advice about this for specific platforms (see, e.g., <https://inside.psych.cf.ac.uk/> sections on EMS and Qualtrics for advice from School of Psychology). The experiment URL will be something like <https://jatos.mindprobe.eu/publix/0000> or <https://run.pavlovvia.org/UserName/experimentName>. To these, you would need to append `?PROLIFIC_PID={{%PROLIFIC_ID%}}` to log a Prolific Id or `?SONA_ID=%SURVEY_CODE%` to log an EMS id.

14.3 PsychoPy Builder to Pavlovian

If you are creating an experiment in PsychoPy Builder, there is built-in functionality to push the paradigm to pavlovian.org. First you would need to have created an account on Pavlovian, and log into your account from PsychoPy Builder.

Pavlovian uses gitlab in the background. Git is extremely useful and worthwhile to learn about. However, if you do not want to fuss with it, it is recommended to ensure that your experiment is in a fairly final stage, especially with respect to the incidental materials it will load (any spreadsheets or audio files your PsychoPy calls) before you initially push it to Pavlovian. This initial push sets up your project's GitLab page, and if you do not want to manually add new files to it, make sure everything is there on the first set-up. Pushing changes you make to your PsychoPy itself is fairly smooth via the Builder interface.

14.4 Anecdote: Instructions for transporting a paradigm from Open Sesame to MindProbe, and recruiting via Prolific

The following instructions are a guide for the general process of collecting data online via mindprobe.eu and Prolific including how to set up the study and download the data set.

14.4.1 Step 1: Exporting the experiment from OpenSesame

The experiment needs to be exported from OpenSesame into a zip file format that can be uploaded to Mindprobe.eu. To do this in OpenSesame open your experiment and go to ‘Menu’ - ‘Tools’ - ‘OSWEB’ and click ‘Export experiment as a JATOS study’. Before doing this, it is worth reading through the required steps for integrating with your experiment recruitment platform at <https://osdoc.cogsci.nl/3.3/manual/osweb/workflow/>. Before starting the process, be sure you have included code needed to log data passed from the participant recruitment system you are using.

Ensure the .osexp file is saved as a .zip file.

14.4.2 Step 2: Importing the experiment into Mindprobe.eu

To upload the experiment into Mindprobe, open Mindprobe and click ‘Import study’ in the top left-hand corner of the browser. Select the .zip file created in Step 1 and click ‘Import.’ The experiment will now appear in the toolbar on the left-hand side of your Mindprobe account.

It is beneficial at this stage to re-name the experiment with a title relevant to the study itself. Click on the experiment and go to ‘Properties’ (blue button located in the middle of the screen) to change the title.

14.4. ANECDOTE: INSTRUCTIONS FOR TRANSPORTING A PARADIGM FROM OPEN SESAM

14.4.3 Step 3: Generating a study link

A link for the experiment needs to be created to be shared to online data collection platforms such as Prolific.

In Mindprobe click the study you wish to generate a link for using the left-hand side toolbar. Click the green box ‘Study links’. This will take you to the ‘Worker & Batch Manager’ page of Mindprobe. On this page you can create study links to send to participants and organise participants into groups named batches to better handle the data.

To generate a link to upload to Prolific select the green box ‘Get study links’ - ‘Choose type’ - ‘General multiple.’

A General Multiple link is the most suitable to use for Prolific as it can be used multiple times in the same browser by different participants.

For more information on the several types of links that can be generated via Mindprobe visit <https://v2.jatos.org/Worker-Types.html#-mturk-sandbox-worker>.

Once the link is generated copy and paste it somewhere you can easily find it. You will need it to recruit participants.

14.4.4 Step 4: Begin setting up the experiment on Prolific

To set up your experiment on Prolific first ensure you are in the correct workspace.

Click ‘New study’ in the top right corner of the screen. This will take you to a window titled ‘Study details’ here you can fill out the relevant information including study title, study details and technical requirements including devices allowed and additional software.

Note - it is important to be as specific as possible within the study details section so that potential participants know exactly what to expect.

14.4.5 Step 5: Recording Prolific Ids

Once the ‘Study details’ section is complete scroll down to the next section titled ‘How to record Prolific IDs.’

Click ‘I’ll use URL parameters’ because you want to upload your study to Prolific using the link you generated in Step 3. In the box above, keep the text that is already in that box but change the ‘&’ after the first ‘studyID’ to a ‘?’. The study will not run successfully without this step.

Next, paste the general multiple link you generated in step 3 in front of the text that is already in the box.

14.4.6 Step 6: Generating a completion URL

Continuing in Prolific, scroll down to the next section titled ‘How to confirm participants have completed your study’

In this section you want to click ‘I’ll redirect them using an URL’ to redirect the participant back to Prolific once they have completed the study via

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MindProbe. This will automatically provide them with a completion code as opposed to manually providing a code for each participant.

Copy the URL in the box and open your Mindrope.eu account.

Once you are back in your MindProbe account, click on the experiment using the left-hand toolbar. Click on the green box ‘Properties’ and paste the prolific completion URL into the box named ‘End Redirect URL.’

14.4.7 Step 7: Finish setting up the study on Prolific

Once step 5 is complete return to Prolific and scroll down to the section titled ‘Audience’.

Enter the number of participants you wish to recruit; this will be equivalent to the number of study spaces that will become available on Prolific. Enter the location you wish to recruit your participants from and which study distribution you intend to use.

Next scroll down to the ‘Pre-screen participants’ section. Here, you can add demographic criteria to pre-screen participants for your study to ensure you reach your intended target audience.

Note - it is important to add your eligibility criteria for your study here for example, target age, gender or need for normal to corrected vision. You can add this information as a reminder in the study details section, but you can’t specify screening criteria only within your study title/description. It is vital to enter eligibility criteria in the pre-screen section.

Next scroll down and add an estimated completion time for your study and

the amount you intend to pay participants for completion. Prolific has a principle of ‘ethical rewards’ and recommends that researchers pay participants £9 an hour. However, the minimum pay is £6 an hour.

When deciding your study cost it is important to remember the higher the payment the more motivated the participant will be to complete your study to a high standard therefore, producing a better quality of data. If you can afford £9 or more per hour, that is better.

14.4.8 Step 8: Publish the experiment to Prolific

At this stage there are 3 options:

1. Save your study as a draft and return to it at a later date when you wish to publish to Prolific.
2. View a preview of your study. You want to ensure you do this before publishing your study to proofread the study description, ensure the link to the study is working as intended and check the data is being correctly stored in Mindprobe.eu.
3. Publish your study. Only do this once you have previewed your study and checked it is storing the data successfully in Mindprobe. It is a good idea to publish it first for a small sample of people to check that the links, re-direction to Prolific, and data recording are all working as intended.

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You can also schedule a date and time for your study to be published by selecting the menu arrow on the ‘Publish’ button and then selecting ‘schedule publish.’

Once the study is published you are able to increase the number of study places. To do so when the study is active press ‘Action’ located next to your studies name then press ‘Increase places.’

14.4.9 Step 9: Downloading your data from Mindprobe

Once participants have completed your experiment on Prolific the data will be recorded on MindProbe.eu. You need to download the data from Mindprobe into a file suitable for analysis.

To complete this step go to your Mindprobe account and click on the experiment using the left-hand side toolbar. Click ‘Study results’. In the middle of the screen, you should see all the data that has been collected with a time and date stamp next to each data set.

A shortcut to view all the finished data is to press ‘Filter Builder’ then ‘Add condition.’ In the box ‘Field’ select ‘State,’ in the box ‘Condition’ select ‘Equals’ and in the box ‘Value’ select ‘Finished’. To finish click the arrow to the right of the boxes. All the finished data sets will now be shown in the middle section of the screen.

If you want to export the finished data only after applying this condition, click ‘Export data’ then ‘selected’. If you want to export all the data collected

on Mindprobe click ‘All’.

Save the data file.

14.4.10 Step 10: Transforming data to .csv file

At this stage, the data saved in step 8 is not in readable format. It needs to be transformed into an excel file or a .csv for analysis.

To complete this step open OpenSesame and click ‘Tools’- ‘OSWeb.’ Then press “Download JATOS file to csv.”

Click on the file and save the file as a .csv for analysis.

14.4.11 Step 11: Granting participant payments

Once you have checked whether a participant’s data has met your inclusion criteria you need to decide whether their submission is to be approved and paid, rejected or returned.

To do this go to your Prolific workspace, click on ‘Project’ on the left-hand side toolbar. Here, you will see all your active, scheduled and completed studies. Click on your study name, this will show all of the participants who have completed your study.

Those who say ‘Returned’ next to their Prolific ID did not complete your study, left early or withdrew their consent, those who say ‘Awaiting review’ are those who need to be paid if there data was collected successfully.

14.4. ANECDOTE: INSTRUCTIONS FOR TRANSPORTING A PARADIGM FROM OPEN SESAM

To approve a participant payment, click the green check mark on the right-hand side of the page next to the participants prolific ID.

Alternatively, click the tick box on the left of the participant's ID and then click the blue box above that says 'Approve'. Using this method you can approve multiple participants at the same time.

If you wish to reject a participants submission click the red cross on the right-hand side next to the participant's prolific ID. This will direct you to a pop-up where you will be asked to explain why you are making a rejection for this participant.

Prolific is extremely strict regarding rejections therefore it is vital to only reject participant submissions if you have a strong reason to believe the participant has been clearly negligent. A link detailing the valid and invalid reasons for rejecting a submission are explained in detail in the following link:

<https://researcher-help.prolific.co/hc/en-gb/articles/360009092394-Approvals-rejections-returns>.

A better alternative to rejecting a participant's submission is to ask them to return their submission. You may wish to do this in cases where there has been technical difficulties and a full set of data was not collected successfully or in cases where a participant withdrew their consent but it says 'Awaiting review' next to their Prolific ID.

To do this you need to send the participant a message politely asking them to return their submission along with your reasoning for this request. To send a message to a participant, click on your project and click the envelope next to their prolific participant ID.

It is important to give participants 7 days to return their submission. If a submission is not returned during this time frame you will need to contact Prolific. Note- do not leave the submission as ‘Awaiting review’ because Prolific automatically approves all unapproved submissions after 21 days.

Next to some participants IDs you may see ‘Timed-out’, if this is the case the participant exceeded the maximum amount of time decided by Prolific based on your studies duration. No further action is usually required for these types of submissions. However, occasionally a participant may message saying they completed the study, if this is the case you need to review their data. These types of submissions can then be manually approved by clicking the green tick next to their Prolific ID.

Chapter 15

Experimental protocols

The experimental protocol is like a recipe for running your experiment. Each experimenter responsible for a discrete data-collection initiative must write the script for completing a run. This script should be sufficiently thorough that a trust-worthy, non-lab-member psychologist could run it correctly from the script alone. Imagine that you experience an emergency shortly before your participant is scheduled to arrive. Your protocol should be detailed enough that you could run into the Tower building, find another Year 3 or PG student, give them your protocol, and they could do everything from setting up the lab to saving the data.

(Note, we would **never** actually do that. Of course non-lab members cannot do our lab work - see data security. But nonetheless your protocol should be detailed enough to allow it.)

15.1 Writing your protocol

The PI will direct you to an appropriate example protocol to work from. Where possible, you may copy from the existing protocol. There no point using your creativity to find new ways to describe instituting similar set-ups. Familiarity also helps other lab members know when procedures were the same and when they differed.

Writing a protocol is an exercise in theory-of-mind. You need to think carefully about what someone else does not know about your procedure, and address each point you can think of explicitly.

Our paradigms will typically include the following sections.

15.1.1 Setting up

Your protocol should start with whatever must be done *before* the first participant arrives. As a general rule, set-up should be complete 10 minutes before the participant is expected. Assume this, and work backwards for determining how early setting up the lab for your procedure must begin.

Set-up will typically include rebooting the computer(s), applying or checking any settings (e.g., for volume, screen refreshing, color temperature, etc.) specific to your experiment, and arranging the workspace. Include every detail that needs to be applied or checked for your experiment.

15.1.2 Greeting and consent

How will your participant find the lab? Will s/he be able to reach it on their own, or do you need to meet them outside? You want the participant to feel welcomed and not stressed about getting there, so plan to do what it takes to ensure they know where to go.

Remember that your participant might never have been in a lab before. They might not know what they may touch, where they may sit, etc. Instruct them on everything - where to sit down, where to place their things, etc. If they bring food or drink in, they may not have it during the session, but you may be able to keep it somewhere safe for them while they are working.

After the participant is settled, the first thing they must do is consent to take part in the study. Consider how you will present consent to them, and how you will emphasize the main points of the consent document.

15.1.3 Instructions and practice

All of our experiments should be designed to guide participants through the task, so that after an initial practice period the experimenter is confident that the participant understands what to do. The best means for doing this will vary depending on your project. Here are some general principles that we have found to be effective:

Do not plan to have your participants read through instructions on their own. Of course they *can* read, but always keep in mind that many participants want to get in and out of the lab as quickly as possible. They will assume

that they can just figure it out, and click through instructions like it's their iPhone legal agreement update. They will get to a trial, and call you over because they do not know when to respond or what button to press.

Either use the on-screen instructions to remind yourself of how you will aurally explain the instructions to the participant, or implement a read-instruction system where the participant cannot advance past a screen on their own. One way to do this is to end the instructions with something like "Now, get the researcher's attention before continuing", without telling participants how to get off the screen (and choosing a non-obvious exit key that the experimenter knows as the only way to advance). This way, you have the opportunity to check that the participant understands what they are about to do before they begin.

Include practice trials, and design them to be representative of experimental trials, but easier. Over-represent the easiest conditions in your practice block so that it will be really obvious to participants what to do. Consider including an accuracy criterion for advancing to the experimental trials, where the participant has to repeat the practice block until they have proven they know how to do the task. (Note: this is more advisable for some paradigms or components than others, and only possible to implement for conditions where you expect very high accuracy.)

Explain in your protocol how to guide participants through the instructions and practice. After you have run some pilot participants, you may want to update this section after seeing what they find difficult.

15.1.4 Monitoring or on-call

For most of our studies, participants complete the experimental trials independently after passing the practice block. There may be elements of participants behavior that the researcher must monitor or record. If so, what are these? How do you make recordings? What decisions will the researcher need to make? Detail these instructions carefully so that another could reproduce exactly what you do to monitor the participant.

If the researcher is simply “on-call”, waiting for the participant to finish, what is the researcher allowed to do? Is it fine to do some quiet work during this time? Use the protocol to let the researcher know that.

For what reasons is the participant likely to need assistance during this time? Is there a new block of trials with different instructions? Are there breaks? How often? Include these details in your protocol so that the researcher knows what to expect. Participants might ask about these details too.

15.1.5 Saving and break-down

When the participant is finished, the researcher should thank them, debrief them (and pay them, if they are working for pay). The researcher should escort the participant out of the secure lab area or building (for the participant’s convenience as well as to comply with safety and security rules). What else needs to happen at the end of a run? Is there anything to be recorded? How should the researcher save the new data? After the final participant of the day, how should the lab be shut down? If the participant is getting EMS

credits, how does one apply those?

15.1.6 Exceptions and unusual events

These typically include what to do if a participant decides to withdraw consent. Participants may withdraw for any reason. They may request to have their data deleted. In case they do, you should detail how one would find the new data, and all the steps needed to show them it has been deleted.

If the participant is taking part for money and withdraws consent, it is only necessary to pay them for the amount of time they spent. Deciding how to pro-rate participant pay can be sensitive. A good general rule is to round up to the nearest quarter hour and apply the advertized hourly rate.

If the participant is taking part for EMS credit, then how to proceed depends on the reason they give. Do ask why they are withdrawing consent, and delete their data if requested. Depending on the reason for withdrawing, their credit might be pro-rated similarly to applying payment. This should always be discussed with the PI.

15.2 Testing your protocol

New protocols must always be tested, and perhaps revised, before beginning your study. If your experiment is part of an existing series, you may be authorized to skip testing because subsequent experiments are likely to be very similar to others in the series. But if your protocol is for a new project, or you are new to the lab, you must take it through the following tests.

First, run through it yourself. Try not to bring any unwritten knowledge to interpreting your protocol. See if there is anything you need to add. You should always do a complete run of your experiment too, to check that it really does what you expect it do, stops when you think it will stop, etc.

Then get another lab member to go through the protocol, and to actually perform the set-up as the protocol instructs. If the lab member cannot do it correctly based on the protocol instructions, then revise the protocol based on their feedback and try again. The lab member may also have suggestions for the remainder of the protocol for you to implement. Run the lab member through your experiment if you have not already pilot-tested it within the lab. The lab member may have suggestions for tweaking your instructions, or may help find uncaught errors or bugs.

Once another lab member believes the protocol is fairly complete and the experiment runs as intended, the protocol and experiment should be shown to the PI. (Note: Depending on how much support the researcher needed to program the experiment, the PI may already have seen the experiment itself by this point.) If the PI finds it complete, then she will authorize you to schedule a naive participant for a time that she (or another senior lab member) can observe. The observing lab member will be present for the set-up, at least the instructions/practice if not the whole run, and break-down to see how the protocol worked and how closely the researcher followed it. After this exercise, the observing lab member and researcher will discuss how the run went, whether any changes need to be made. If no changes need to be made, then the run may be considered the first participant in the study. This participant number may need to be marked for exclusion if major changes

are requested. This decision should be explicit, and recorded in the notes the researcher keeps about participant runs.

15.3 Clearance to begin

Once the researcher has completed a supervised run, and the supervising lab member finds the run successful, with no changes to implement, then the researcher is cleared to begin recruiting the necessary number of participants.

Chapter 16

Recruiting participants

16.1 Adults from the university community

16.1.1 Experiment Management System (EMS)

The most convenient way to recruit adults capable of consenting on their own behalf for your research project is to use Cardiff University's Experiment Management System (EMS). EMS has been purchased by the School of Psychology (it is actually more widely known as SONA systems) to manage and monitor study sign-up procedures in conjunction with the participant panel scheme. Year 1 and Year 2 students on psychology degrees are required to participate in research as part of their education, and researchers in the School are responsible for making opportunities for this participation available. Participants can sign up for studies online, researchers can set up their studies online, and administrators can ensure participants have completed all

their requirements. This is all done from any web browser, and is available 24 hours a day.

EMS is convenient for researchers for several reasons. Researchers can easily signal when they would like to run sessions by offering a schedule of appointments to the available participants. Participants then claim the session they prefer. Settings within EMS allow researchers to exclude participants based on stated criteria or based on studies they have completed in the past, minimizing time spent working with ineligible participants and making it unnecessary to keep logs of previous participants identities. Researchers may also offer study opportunities for “credit” (meaning that students can earn the course credits they need for taking part) or alternatively for pay. The same study can be offered for either credit or payment.

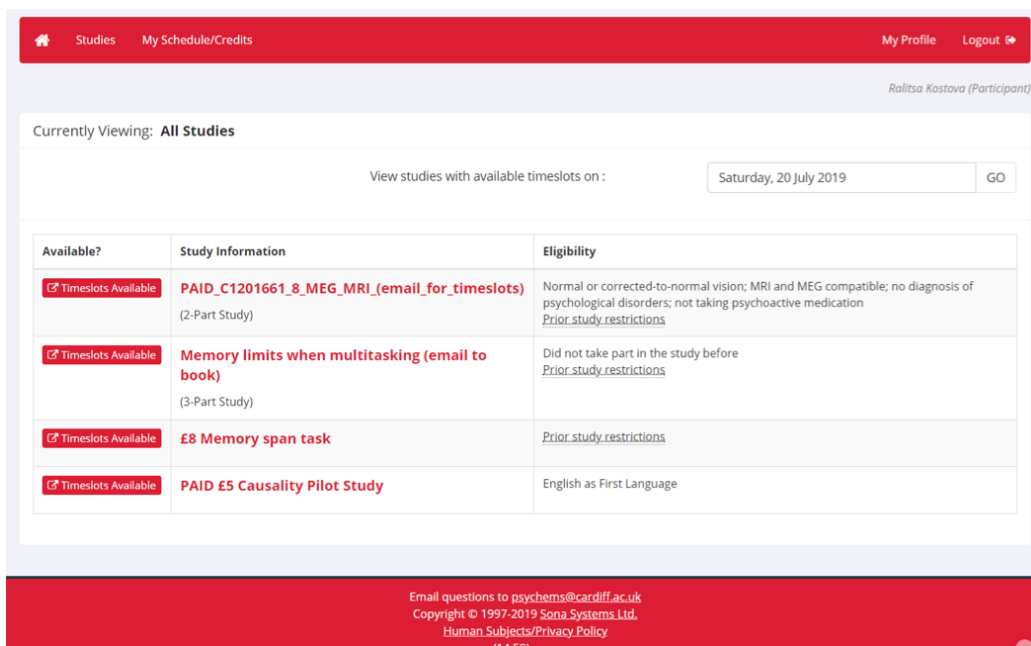
16.1.2 Access to EMS

The first thing to do is to obtain a researcher account from the EMS team. This can be done by emailing psychems@cardiff.ac.uk. Just let them know that you are a researcher, working with Dr Candice Morey, and you need an account to post an experiment and shortly they will send you an email with your credentials.

If you are completing your undergraduate BPS project, there is a ethics training process that you must complete before accessing EMS. Instead of emailing to request an account, your EMS account will be granted upon completion of this mandatory training.

16.1.3 Administering an experiment on EMS

This is quite straightforward: when you create a new study, you will be prompted to fill in fields for all the information needed to advertize a study. Please refer to the detailed EMS user guide linked on the EMS log in page (note - this is an internal document accessible only to Cardiff staff and students). There are strict rules for naming your study so as not to bias students in favor or against choosing your study. Available studies are listed to students in a random order. Note that participants can only see the calendar with available studies on the selected day (Fig. 1). Participants see whichever studies have availability on the day they select, not all studies.



The screenshot shows the EMS interface for a participant named Ralitsa Kostova. The top navigation bar includes links for Studies, My Schedule/Credits, My Profile, and Logout. The main content area displays 'Currently Viewing: All Studies' and a filter for 'View studies with available timeslots on: Saturday, 20 July 2019'. Below this is a table of available studies.

Available?	Study Information	Eligibility
Timeslots Available	PAID_C1201661_8_MEG_MRI (email_for_timeslots) (2-Part Study)	Normal or corrected-to-normal vision; MRI and MEG compatible; no diagnosis of psychological disorders; not taking psychoactive medication Prior study restrictions
Timeslots Available	Memory limits when multitasking (email to book) (3-Part Study)	Did not take part in the study before Prior study restrictions
Timeslots Available	£8 Memory span task	Prior study restrictions
Timeslots Available	PAID £5 Causality Pilot Study	English as First Language

The footer contains contact information: Email questions to psychems@cardiff.ac.uk, Copyright © 1997-2019 Sona Systems Ltd., and a link to Human Subjects/Privacy Policy.

This means that if you do not open any timeslots, registered participants won't be able to see your study at all. The system is designed to minimize back-and-forth between participants and researchers scheduling them. You

say when you want to work, and participants either take the appointment or leave it. This works smoothly when there is a large population of participants available. If for some reason you need to personally arrange appointments (e.g., if you are running a multi-session study and need to negotiate session times per participant within certain time frames) you can still use the system to recruit. In this case, you would need to create two studies on the system and condition participating in the second study on having completed the first.

Once you assign timeslots you can also change them, delete them, etc. Just as researchers expect courtesy from participants regarding cancelling and rescheduling, researchers must extend similar courtesy to participants. If you are offering appointments for credit, researchers and participants may cancel for any reason if they cancel 24 hours or more from the appointment time. If you must cancel a session inside 24 hours, then you must grant the participants credit. If a participant fails to show up for a credit appointment, the researcher has some discretion over what to do about it. The researcher is entitled to issue a penalty for this, and should do that unless the participant has contacted the researcher with a satisfactory reason for missing the appointment or cancelling on short notice. Assigning credits and penalties is done through EMS. Once the appointment timeslot is in the past, a dialogue for that appointment becomes available, allowing the researcher to choose whether 1) credit should be assigned, 2) the appointment was a no-show (excused), or 3) the appointment was a no-show (unexcused). Choose whichever option matches the circumstance.

With paid participants, you similarly need to cancel 24 hours ahead of time,

otherwise the participant is entitled to payment.

EMS will remind participants by email of their appointments. You can set your experiment to alert you (or not) when there are sign-ups and cancellations, or when you have filled appointments coming up.

16.1.4 Alternatives to EMS

Sufficient numbers of credit and paid participants are available via EMS during both academic terms. During holidays and exam periods, virtually no credit participants are available and the population of paid participants diminishes. If you need credit participants, you must plan to recruit during term time.

Even when paying participants, EMS may not yield sufficient numbers during the summer months. Alternative methods of recruitment during the summer to complement EMS include advertising paid research opportunities on social media and flyering in locations where eligible participants are likely to notice (around campus, in cafes, etc.). Calls for participants may be posted on the *Take part in research* Yammer channel.

16.1.5 Paying participants

Paying participants is essential for collecting data outside of term time, but of course it is only an option if you have a source of funding for it. Payments for participants are administered by the researcher, with funds released to

the researcher by School of Psychology Finance when an appropriate budget is available to draw upon.

Do you have access to a participant payment budget and want to proceed? You must request the funds you need from Psychology Finance (psychfinance@cardiff.ac.uk) several days before you need the money. It can take up to 2 weeks to set up, so plan this well in advance. Psychology Finance need to know the total amount you need, what university account to draw it from. There is a template spreadsheet researchers must use to keep track of who has participated and how much they should be paid. Researchers must give participants who completed the study the link to Finance's Qualtrics survey, which is used to collect participants' bank information. Once per week, the contents of this online survey are married up to the ongoing spreadsheet submitted by the researcher, and participants listed in both are issued the payments.

16.2 Children from the local community

Sometimes we recruit children to take part in our research. We have recruited children by arranging to test in schools and by inviting families to bring their children to the university for testing.

Regardless of how you are recruiting children, you will first need to acquire a basic DBS (disclosure and barring service) check. This is an authorized confirmation that you do not have a criminal record that prohibits working with children or other vulnerable populations.

16.2.1 How to recruit via schools

It is necessary to first make contact with the head teacher (or sometimes a designated community liaison) of a school and explain 1) what your study entails for the participants, and 2) what we anticipate learning from it. The head teacher can authorize us to send consent forms for parents home with children, and make a quiet place in the school available for testing. In the first instance, contact head teachers via email to the school's general account or via hand-written note, sent in the mail. Start this process as far ahead of testing as you can; it can take a while for interested schools to make contact with you.

Head teachers are extremely busy - we cannot expect that assisting us with research is their top priority. We must make ourselves available on their timetable and respect the limits (in terms of time in the school, etc.) they are comfortable making available to us. Note that our ethics approval *requires* explicit, opt-in consent from parents, so the first step is always to send forms home explaining what we are doing and why we think it will be beneficial. We can only test children for whom we receive completed, signed parental consent forms.

We have only begun working in schools, so cannot yet describe in detail our means of feeding results back to them. We have offered to put on workshops, targeted either toward staff or students, to teach principles of memory.

There is a record of contacts attempted with local schools maintained by CUCHDS. If you are attempting to establish a relationship with a school, you should first consult this record to learn whether contact has been attempted

before and whether the school is currently open to working with researchers.

16.2.2 How to recruit via the CUCHDS database for on-campus testing

Often, it is necessary to test children in a controlled setting. We use the CUCHDS facilities for this (further details on the facilities at CUCHDS are available [here](#) or [here](#). CUCHDS features several child-friendly testing spaces, including some equipped with EEG, fNIRS, and eye-tracking capabilities, bookable by personnel with access (the PI can request access on your behalf - this is done via psych-facilities.) For most lab spaces it is also important to gain access to the Google Calendars for these rooms. These Google Calendars are controlled by Dr. Ross Vanderwert; contact him to request access (a Gmail/Google account is required). For safety purposes and to allow others to book testing space, if a testing session is no longer going ahead, make sure to delete entries in all calendars and inform anyone who was going to be onsite/available for supervising/oversight that the session is no longer going ahead.

During busy periods lab space can become difficult to book. Make sure that you have communicated with other research groups before recruiting families to take part in your research to check if there are any considerations you should be aware of. For example, someone might have a deadline for recruitment which is close, and they might want priority over the lab spaces. Such decisions as to who has priority should be discussed between relevant lab members as well as PIs to make sure that the spaces are being used

in a fair and equal manner. Another such example is that research group A might want to collect adult data using the undergraduate student population (but using the same equipment as a child sample they have collected) and research group B is collecting child data during term time after the school day has ended. In this case it would be good practice for research group A to ensure that they have finished any testing sessions for the day by 15:00 in the afternoon, so that research group B may use the same lab space from 15:00 onwards when children would be more available. Similar considerations should be made during school holidays, i.e. lab space should ideally not be used with adult participants during school holidays to help ensure as many child participants can be booked in during periods when they are available.

CUCHDS maintains a database of families (REDCap database, also known as the Junior Scientist Database) who have indicated that they would like to take part in research projects (see TinyToTots). Access to this database is controlled by Dr. Ross Vanderwert and Dr. Sarah Gerson. Training on how to use the participant database is required before you can contact families. Dr. Sarah Gerson or Dr. Ross Vanderwert can advise on who should be contacted to arrange a suitable time for training on how to use the database.

When contacting participants, it is important to greet them personally. Do not send a bulk email to everyone in the database that meets your inclusion criteria. Send a personal message, offering a chance to participate. Make sure to check that the families are interested in the mode of research you are conducting before making contact. Some families only wish to take part in online studies, some in person, and other are happy to take part in both modes of research. If the family has indicated a preferred means of contact

other than email, use the preferred means instead. Include details about what the child would be doing, how long it will take, who is eligible, and general information about what we are studying. It is important to note that some families remain on the database but since signing up have requested not to be contacted about further research. Make sure to check that they are interested in being contacted about future research and also have given us permission to contact them via at least one method. DO NOT contact families who have not given us permission to contact them and do not want to hear about future research.

When emailing families, it is useful to use the lab email address (workingmemorylab@cardiff.ac.uk; for access talk to the PI). This helps to ensure that recruitment can be maintained across the lab even when lab members change over (this is highly likely during the summer months, which is a key recruitment time for developmental testing, when undergraduate placement students are likely to end their time in the lab and summer placement students join the lab), aids in timely responses to families when they email as multiple people can check the email inbox, plus helps to keep your own work email from being clogged up with recruitment emails. It is also good practice to ensure that the emails to families are checked at least twice per day, once in the morning and once in the afternoon. This is to ensure that families have a good experience with researchers. Similarly, families might be more apprehensive of allowing their children to take part in research studies, especially those which use equipment such as EEG and fNIRS, than adult participants. Parents might be concerned that their child will be uncomfortable or could be harmed in some way. Ensuring

that families have a good experience when helping us with our research is important as we are not the only research team contacting these families and if a family has a bad experience this could damage the reputation of the whole of CUCHDS research teams, not just yourself. Additionally, families who have had good experiences in helping with research in CUCHDS often come back year after year to take part in more research studies and are keen to help out where they can, so maintaining this community of families interested in helping with research is vital.

It is important that the same families are not sent emails too frequently, both from the same research group, but also from different research groups within CUCHDS. It is important therefore that before recruitment begins, using the participant database, that contact is made with other research groups in CUCHDS to see if they are also recruiting families to take part in research. It is recommended that researchers team up when recruiting families, especially during busy times such as school holidays, so that as few emails are sent as possible. For example, if researcher A is looking to recruit children between the ages of 7-10 years of age and researcher B is looking to recruit children between the ages of 4-6 years of age, it is recommended that both researchers sit down to draft an email template, detailing both studies and criteria, which can then be sent to families who have two or more children in this age range. Remember not to put too much information in these emails and to instead give a summary of the important information and attach the information sheets for the parents to read. By combining emails this reduces the number of emails that families are receiving and also aids in booking in sessions. Researcher A and Researcher B should also agree on times when they could

hold siblings sessions so that the family would only need to come into the lab for one visit (depending on whether the study requires multiple visits or not).

Similarly, it is good practice that when two or more research groups are recruiting children in the same age bracket, that the researcher who meets the family first enquires if the family would like to take part in more studies. If the family indicates that they are interested in taking part in more studies, the researcher should make a note of the child's REDCap ID number (the number from the participant database) and send an email to the other researcher with the child's REDCap ID number and inform them that the family are happy to be contacted about the study. DO NOT send personal information about a child or the family in an email, this is in case the email is opened by someone other than the indented recipient or is sent in error to the wrong email address. Always refer to children and their families with regards to anonymised information such as the REDCap ID.

In order to give back to the CUCHDS research community, it is good practice when recruiting families from outside of the CUCHDS database to encourage families who are interested in taking part in more research studies to fill in the database registration form (<https://tinyurl.com/T2TDatabase>). This link can be added to emails, captions of recruitment posts on social media, recruitment fliers, etc. It can also be used to generate a QR code if you are adding the link to paper documents/fliers so that parents can scan the QR code and be sent straight to the sign-up link.

For further details on how to use the participant database contact the PI

for access to the Junior Scientist Teams channel where the CUCHDS Booking and Recruitment document, and other training materials can be found. These documents include sample email templates to parents as well as how to log communication with families to ensure that families are not contacted too often in a short space of time.

16.2.3 Recruiting via social media

Another way to recruit families is via social media. TinyToTots maintains a Facebook page, and can post a call for participants for a particular study opportunity. To make the most of this, you should create an image including the relevant details that can be posted to Facebook. Details should include what the participant will be doing, who is eligible, how long it will take, until when you are recruiting, and contact information. The flier should include a fun, colorful image to attract attention. The Facebook account is managed by Dr. Sarah Gerson.

The *Participate in research* Yammer channel may also yield some recruits. It is monitored by a lot of university staff, who actively participate themselves and might be open to bringing their family in for research opportunities.

Chapter 17

Dealing with the unexpected during data collection

We design experimental materials to be easily administered and as fail-safe as possible. However, that doesn't mean that nothing can go wrong during your data collection. Here is a description of how participant running usually goes, and then anecdotal exceptions that have occurred to us more than once.

17.1 The slow participant

Most student participants are not new to taking part in research, and understand the drill. They want to get in and out of a session as quickly as possible. This is so typical that we must account for it in the design of experiments and ensure that participants learn that they will not benefit from random responding. However, some participants take much longer than is

typical to respond. Sometimes, we never learn why, we simply see that their session takes a long time and their response speeds are much slower than average. Sometimes, these participants wish to discuss how they are doing the task in detail with the experimenter. If a participant wants to discuss the experiment with you, always engage with them as far as possible (up to explaining the hypothesis, which you can do after they finish).

17.2 The inattentive participant

Remember how most student participants have learned that it is in their best interest to finish as quickly as possible? Some will try to speed everything along: they will grab the mouse and start trying to enter information without knowing what they are doing, will start mashing SPACE, ENTER and random keys to get the study going. Data from this behavior would not signify anything. This is the reason our policy is always to personally explain instructions and have a supervised practice session. Practice sessions should be designed so that the experimenter is certain the participant understands the task, and that the participant is not inadvertently rewarded for speeding through examples and instructions.

Sometimes we cannot ensure that a participant will not choose to randomly respond after they are left on their own. Our tasks are often hard, and it would not always be reasonable to impose an accuracy criterion (though sometimes it is). Our strategy is to ensure that they know how to do the task, not to set it up in such a way that non-responding would allow them to finish earlier, and to build in some trials that are meant to detect insincere

attempts. Usually, this means including some very easy condition. Even if these trials are not meant to be part of the main design, they can be used to check that the participant understood what to do and was not responding randomly.

17.3 Sleepy, sick, or inebriated participants

Student life is difficult, and sometimes you will greet a participant who isn't in fit condition to be working on anything. The PI has seen participants who came over directly from the pub, who were so ill that they ought to have been in bed, or who were so tired they could hardly sit up. These participants are trying too hard to meet their requirements. If you encounter a participant in this state, the best option is to offer to reschedule them. They will not learn anything from trying to take part.

17.4 The participant with a lot of questions

Occasionally, a participant will want to know a lot of detail about how the study works or what will be done with the data. Intellectual curiosity is always to be encouraged. Engage with questions as best you can, but if you do not feel up to answering all of them, refer the participant to the PI or another senior, responsible researcher.

Often these participants want to see their own data. Our general policy is only to release anonymized data sets. Note down this participant's email

address so that we can alert him or her when the anonymized data are available. In some cases, we may be okay with giving the participant their own data set as well. The reason that we generally do not do this is that in our experimental tasks, usually there is little (if anything) to be learned from one person's performance. There is nothing diagnostic we would be able to say about certain task scores, and there is some worry that a participant might try to read too much into their performance. Giving access to the anonymized data would be sufficient to let them have a go at data analysis and satisfy their curiosity about the results.

Chapter 18

Tips and tricks for working with children

So you want to run a developmental study and work with children. There are several things which need to be considered when designing and conducting a developmental study.

18.1 Designing an experiment for children

Firstly, all developmental studies should be given a cover story and gamified to the extent possible. Unlike adult participants who will tolerate (to some extent) a task in which they are just instructed what to do, children have a much lower attention threshold and will quickly get bored of completing a task if all they are given to look at is a white screen with some black shapes (for example). It is worth noting that children will have a shorter attention

span than adults, and younger children will have an even shorter attention span than older children and that their attention span will be decreasing through every aspect of the session (e.g., any preliminary set-up or questions), so it is important that there is also a limited delay in starting the essential tasks you need them for.

When designing a developmental experimental paradigm consideration should be given to the types of equipment to be used during the experiment given what you are thinking of investigating. For example, many primary school aged children are now commonly exposed to touchscreen tablets and may not have used a computer mouse before, so if you are thinking of investigating reaction times it would be important to have children respond on a touchscreen, as they may struggle to answer promptly using a computer mouse, thus invalidating the response times calculated. CUCHDS has a wide range of furniture and equipment which is suitable for children to use and it is recommend to discuss with your PI what equipment may be available for you to use.

The length of study should be as short as possible, whilst also enabling the collection of the minimum amount of required data. It is useful when designing the study to work out what the minimum number of trials to be conducted for each condition should be and then add some additional trials to the task as it is likely that trials will need to be excluded during data analysis because children will often move around/get distracted whilst completing the task, potentially leading to data which is messy and does not accurately represent the behavior you are investigating. Therefore having spare trials can help to mitigate the amount of data that may be lost. Addi-

tionally, putting the most important task for the research at the start of the experimental session is wise to ensure that the children are most attentive and engaged when completing the task. Tasks which are of lower importance should be placed at the end of the session when the children are more likely to be bored and more easily distracted or inattentive. To make it worthwhile for parents to come to the CUCHDS centre most studies for children aged 5-years-old and older tend to last around one hour, including the signing of consent forms and breaks, and most studies will not last over two hours maximum as children will not typically concentrate for longer than this (studies over one hour will often include neurophysiological measures which take time to set up with the child, e.g. EEG capping).

When planning the session structure it's useful to decide on points during the experiment when breaks can be taken to limit the amount of disruption to the task. Children should be informed before the session that they can take breaks and should let the research know when they would like to take a break. When break points are reached during the experiment session the researcher should inform the child that they can take a break if they would like to or continue to play the games. It is often helpful to plan for breaks when switching between tasks and not during a task to minimize disruption during a task, unless the child is completing one continuous task throughout the session.

To help with continued engagement in the task, it can be useful to provide the children with praise and encouragement. For example, informing them every few trials that they are 'doing well', doing a 'good job' etc. If the children believe they are doing well at the game this can help them to concentrate

and continue to try their best. Another form of encouragement is giving the child a sticker when milestones are achieved during the session, e.g. after completing one game and before moving onto the next game.

As a thank you for taking part the children are given a prize (e.g., a small toy), stickers and a certificate of participation. It is also possible to give a small cash honorarium to the parents for taking part as well, but this is often only the case when the parent is required to also take part in the research as a participant or in infant studies. It is also possible to cover transportation costs for parents to travel to CUCHDS, but it is common for most studies to only offer free parking outside of CUCHDS for the duration of the study session for families traveling by car.

18.2 On the day considerations

Some elements to consider when conducting research with children is whether the parent will be required to sit in the experiment room throughout the session. Sometimes (especially during school holidays) parents will bring siblings along to the testing sessions as they don't have other childcare arrangements or they have something else planned for after the session. It is important to consider this when inviting children to take part if the parent needs to be present with the child during the experimental task. If the parent needs to be present in the research room throughout the session, then another member of the research team should be available to sit in the waiting area with the sibling not taking part to help ensure that the participating child is not distracted by their sibling. Another option is to ask both parents (or for an-

other adult) to attend the session so that one adult can wait in the waiting area, however this will make it more difficult to book in child participants as it can be difficult for families to facilitate this. When inviting parents it is good practice to let them know whether a parent/adult will be required to stay with the child partaking in the research or whether they will wait in the waiting area so that families know in advance of the session what is required.

For some families and children taking part in research might be the first time they have visited a university campus and as such it is important to remember that children and parents may be nervous or anxious when attending. As such, it is key that the researchers present themselves in a friendly manner and take time to explain what is going to happen during the research session and what the different rooms are when moving families around the building. One way in which the children and parents anxiety can be alleviated is by showing the child and parent where the research will be conducted (i.e. in which testing room) and how close it is to the waiting area. It may also be advisable to familiarise the child and parent with the equipment, which is going to be used in the research, especially if the equipment being used is likely to be something which they may not have come across before e.g. EEG cap and electrodes, eye tracker, etc.

It is advisable that for older children (5-years-old and up) that parents are asked to wait in the waiting area or observation room during the testing session so that the child is not distracted. Sometimes children will look to their parents for help or get distracted by the parent being in the room, which could affect the data. However, sometimes the child and/or parent is not comfortable with this and will request for the parent to stay in the room

during the research session. If this is the case make sure that there is a chair available for the parent and ask the parent not to distract or help the child complete the task as we want to investigate how the child completes the task. The child should also be reminded that the parent is not able to help them complete the task and that we want to see how well they (the child) can play the games.

Chapter 19

Organizing your research data

It is very important to start the data curation process as you prepare to close data collection for your project. This helps to ensure that no data are lost, that any paperwork and digital data are consistent with each other, and crucially, that the PI knows what materials are available and how they have been archived.

19.1 Is data collection closed?

Because many of the analyses we carry out are based on Bayesian statistics, we do not always need to recruit to a particular, pre-registered N. So in some cases we may stop-and-restart data collection for a particular project. At each of these potential stopping moments though, it is important to log what data we have, what decisions we have made about it, and archive consent forms. When data collection is rolling and managed by one constant

person, we might take a less formal approach. But usually, the close of data collection, even if it might be restarted again, should be accompanied by a **Data Round-up Meeting**.

19.2 The Data Round-up Meeting

It's often the case that the person collecting the data is not the only person analyzing it. But the person responsible for data collection is the only person who knows details about how each participants' sessions progressed, whether anything strange happened during a session that merits exclusion, and where the physical and digital data are. We do not want to lose this knowledge. The Data Round-up Meeting ensures that we preserve important information the data collector has as soon as possible after data collection is over.

Everyone involved in data collection should attend the Data Round-up Meeting, because we will at the same time make decisions about participant exclusions. Everyone who collected data should bring their lab notebook. If a face-to-face meeting including all personnel is impossible, data collectors should at least make their lab notebook available to the PI for this meeting, and have prepared a summary of which participants they think must be left out and why.

19.2.1 Preparing for the Data Round-up Meeting

Everyone attending the meeting should prepare in advance, so that it can progress smoothly. If you were responsible for data collection, you must:

1. Gather all paperwork participants completed and bring it to the meeting. If any data was recorded on paper, there should be instructions for putting it into a digital format. This should also be done ahead of the meeting.
2. Scan all consent forms and send the .pdfs to the PI.
3. Make sure all the data you collected or coded is saved to its designated directory on *labBackUp_cmcorey*.
4. If you are analyzing data in software other than R, think about what output you need from the raw data set, and what form it should take.

All of these actions must be finished before we can begin.

19.2.2 Pull all the data together

First, the team shall compile the data associated with the project from the *labBackUp_cmcorey* folder. It is essential that this folder is up-to-date ahead of the meeting - data collectors, please ensure that you have copied all data to the back-up ahead of time.

At the meeting, we will use this code to compile each individual backed-up data file into a single data frame, called *d* in this example:

```
d = map_df(list.files(path = "", full.names = TRUE),
           read_csv, col_names = T)
```

This code creates a dataframe called *d* that includes all the .csv files in the project directory, which will be entered into the *path* argument.

19.2.3 Anonymization

We shall then examine *d*'s headings for information that could identify the participant. Anything that has to do with the date and time of the session must be omitted. Recorded birthdates must be omitted. Usually, we would not have recorded anything more personal than that, but if so, it must be omitted at this stage.

Obviously, we do need the participants' ages to describe our sample. We can calculate that. First, it is a good idea to first look at the entries for any year-of-birth or birthdate variable to make sure they are plausible.

```
unique(d$YoB)
```

This code will produce a list of each unique value at YoB in the data frame. Check that they all seem plausible. If they are not, consider whether a weird value might have been a typo by figuring out who interacted with that participant and consulting the relevant notebook. The people in the room must reach an agreement on what to do in the event of a weird value. Options include: considering the age missing for that participant, omitting the participant if they fall outside the recruited age range, judging that a transposition was entered (e.g., entering a YoB of 2020 rather than 2002 was an error because the participant was definitely an adult, or that a 2-digit value like 98 actually meant 1998). In cases where the typo seems obvious, we can decide to fix it, but this must be documented in the script.

If we recorded *Year of Birth*, we can now create a new variable converting birth year to age at the time of the study. For most purposes, this does not

need to be very precise. We can do this with a simple *mutate*.

```
d = d %>% mutate(age = 2018 - YoB) # current year - column name containing the birth
```

If we recorded the full birthdate, we probably need age in more precision. The R package *lubridate* is useful for sorting out birthdates and calculating precise ages from birthdates and run dates.

Once we have created whatever new, anonymous variables we need from the potentially identifying data, we can create a new data set with the identifiable data omitted. We would do that with script like this, using the *select* function:

```
# Remove columns that contain ids or timestamps (times could limit the pool to spec
symmsp = symmsp %>% select(-c(Clock.Information, Email, SessionDate, SessionTime)) # TH
```

Include all the variables you want to omit within the `-c()` list. The resulting data frame will include everything except those variables. View the resulting data frame to check that this indeed worked as intended.

This new data frame is the one we will analyze. Save it now to the PI's local directory, and also upload it to the OSF page for this project so all of the lab personnel who need to use it can access it.

```
# Save it - this is the version that goes on OSF
write_csv(symmsp, "symSpan_RuG_anon.csv")
```

We may occasionally need the identifiable data, particularly the timestamps in it. If we need to prove that the data were collected after a certain date,

these timestamps are strong evidence. But the original data need to be kept only in a secure place that is never accessible publicly and that is regularly automatically backed up. Currently the only available spaces that meet this requirement are the Shared and Home drives. Our policy: the compiled, non-anonymized data are stored on Candice Morey's Home drive. She will store the script we write to anonymize the data and the original data on a sub-directory of My Documents/dataSets_cardiff on her Home Drive. But ideally, after the Data Round-up Meeting, the data do not need to be re-compiled again, and all analysts have the same document to work from via OSF.

19.2.4 Participant exclusions

We need to confirm that all of the data we just rounded-up are valid for analysis. Before looking at individuals' performance, we may already know that some participants must be excluded. Depending on the study, we might have recorded responses to eligibility criteria in the data, or may have this information recorded on paper. At this point, we go through this information and compile a list of participant id numbers to leave out of analyses. We want to compile this list as early as possible in the data analysis script, so that any sub-selecting we do occurs to a data frame that already excludes ineligible participants. We also want to keep documentation of why each participant was omitted. We do this in an R markdown notebook. Here is an example code chunk documenting reasons for exclusion and carrying out the exclusions:

```

# The participants we are excluding based on inability to label one or more picture.

# Checking your lab's log or notes, describe any participant number who should be excluded
# 37 was videoed but mouth was not visible (and there was no coder in real-time)
# 54: program malfunction
# 061: experimenter (CCM) accidentally modelled speech rehearsal during practice
# 095, 96, 97 were siblings or friends who were run, but not members of the intended study
# Q: Actually 96 is 6 years-old - can we include, or was there another problem? A: No

# Put all the participant ids who are to be excluded inside this list
exSubs = c("37", "007", "015", "018", "044", "54", "061", "095", "96", "97")

d = d %>% filter(!(participant_id %in% (exSubs)))

# Note excluded participants ages for inclusion in Table B
# 007 - 5 years
# 015 - 6 years
# 018 - 6 years
# 044 - 6 years
# 54 - 5 years
# 061 - 6 years
# 37 - 6 years
# Totals:
# 5 years: 2
# 6 years: 5

```

The object *exSubs* lists each participant id to be omitted. It is important that this list reflects how the data were recorded. We should check at this point whether the participant ids were recorded as characters (and should therefore be inside “ “s) or as numbers (no quotes). When they are characters, we should confirm that each participant id is unique and always recorded with the exact same character string (and that that character string is the one we put in *exSubs*).

The *filter* function performed on data frame data instructs R to keep all rows where the participant id does not match one from the *exSubs* list.

We will also conduct any performance-based checks that may be needed at this point, and add participants who do not meet them to the *exSubs* list and run the filtering step.

19.2.5 Data wrangling

For lab members who are not using R for their analysis, we will discuss the form in which data are needed for your analysis. What variables do you need? Are there dependent variables that need to be calculated? Is any additional filtering (e.g., excluding fast or slow RTs) required? Should it be in long or wide format? To ensure that you exclude the same data that we agreed should be excluded in this meeting, we shall output the data for your analysis from the working R script.

19.2.6 Milestone: Data posted on OSF

At the end of this meeting, the anonymized data file that we begin all scripted analyses with should be posted to OSF. This helps to prevent inadvertent data loss, and means we start all analyses with an identical data file. The exclusion script should be made available to everyone as well. Any R analyses starting from the anonymized file should preserve the participant exclusion code agreed in the meeting.

Uittenhove, K., Jeanneret, S., & Vergauwe, E. (2023). From Lab-Testing to Web-Testing in Cognitive Research: Who You Test is More Important than how You Test. *Journal of Cognition*, 6(1), Article 1. <https://doi.org/10.5334/joc.259>